

BRIDGE COURSE

GRADE – 8

Mathematics

Grade 7

Grade 8



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1. NUMBER SYSTEMS AND TYPES OF NUMBERS

1.1 Natural Numbers

Natural numbers are the counting numbers we use every day. They start at 1 and go on infinitely.

$$N = \{1, 2, 3, 4, 5, 6, \dots\}$$

- Smallest natural number: 1
- No largest natural number — the set is infinite
- Zero (0) is NOT a natural number
- All natural numbers are positive

1.2 Whole Numbers

Whole numbers are natural numbers plus zero. Every natural number is a whole number, but 0 is not a natural number.

$$W = \{0, 1, 2, 3, 4, 5, \dots\}$$

- Smallest whole number: 0
- N is a subset of W — every natural number is also a whole number
- Additive identity: $a + 0 = a$ for any whole number a

1.3 Integers

Integers extend whole numbers by including negative numbers. The set stretches infinitely in both directions.

$$Z = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$$

- Positive integers: 1, 2, 3, ... (same as natural numbers)
- Negative integers: -1, -2, -3, ...
- Zero (0) is an integer — it is neither positive nor negative
- W is a subset of Z — every whole number is an integer

Examples

(i) **Is every whole number a rational number? Give a reason.**

Solution: Yes. Any whole number n can be written as $n/1$, which is in the form p/q with q not equal to 0.

Example: $7 = \frac{7}{1}, 0 = \frac{0}{1}$. So, every whole number is rational.

(a) Addition and subtraction of integer

We subtract the larger number from the smaller number and put a negative sign before the difference so obtained. That is smaller natural number - Larger natural number [Larger natural number Smaller natural number]

Examples

(i) **Simplify: $-465 - 739$**

Solution: To add two negative numbers, we add the numbers without the sign and then we put the negative sign (common sign) before the sum so obtained.

$$-465 - 739 = -[465 + 739] = -1204$$

(b) Multiplication of Integers (Rule)

$(+) \times (+) = +$	Positive $\times$ Positive = Positive
$(-) \times (-) = +$	Negative $\times$ Negative = Positive
$(-) \times (+) = -$	Negative $\times$ Positive = Negative
$(+) \times (-) = -$	Positive $\times$ Negative = Negative

Examples

(i) **Multiply -9 and -13**

Solution: $(-9) \times (-13) = 117$ [$\because (-ve) \times (-ve) = (+ve)$]

(ii) **Evaluate:**

(a) -15×10

(b) $10 \times (-12)$

(c) $5 \times (-13)$

(d) $11 \times (-5)$

Solution: (a) $-15 \times 10 = -150$

(b) $10 \times (-12) = -120$

(c) $5 \times (-13) = -65$

(d) $11 \times (-5) = -55$

(c) Division of Integers

Division is the reverse process of multiplication.

- The quotient of two integers involving two like signs is positive or $(+) \div (+) = +$ and $(-) \div (-) = +$
- The quotient of two integers having opposite signs is negative or $(+) \div (-) = -$ and $(-) \div (+) = -$

Examples

(i) **The product of two integers is -150 . If one number is -30 , what is the other?**

Solution: Let the other number be 'a'. Then, according to question

$$a \times (-30) = -150$$

$$a = -150 \div (-30) = 5$$

Properties of Integers

Property Name	Operations on Integers			
	Addition	Subtraction	Multiplication	Division*
Closure	Yes	Yes	Yes	No
Commutative	Yes	No	Yes	No
Associative	Yes	No	Yes	No
Distributive	$a \times (b + c) = ab + ac$	$a \times (b - c) = ab - ac$	Not applicable	Not applicable

[Where a, b and c should be a non-zero integer.]

1.4 Place Value, Face Value and Expanded Form
(a) Face Value

The face value of a digit is simply the digit itself, regardless of its position in the number.

- In 4,752: the face value of 7 is 7, the face value of 5 is 5
- Face value never changes, it is always just the digit

(b) Place Value

The place value of a digit is the actual value it represents based on its position in the number.

Place Value = Face Value $\times$ Position Value

Position	Position Value	Digit Example	Place Value
Ones	1	8 at ones	$8 \times 1 = 8$
Tens	10	8 at tens	$8 \times 10 = 80$
Hundreds	100	8 at hundreds	$8 \times 100 = 800$
Thousands	1,000	8 at thousands	$8 \times 1,000 = 8,000$
Ten Thousands	10,000	8 at ten-thou.	$8 \times 10,000 = 80,000$
Lakhs	1,00,000	8 at lakhs	$8 \times 1,00,000 = 8,00,000$

(c) Expanded Form

The expanded form of a number expresses it as the sum of the place values of all its digits.

• $3,579 = 3 \times 1000 + 5 \times 100 + 7 \times 10 + 9 \times 1$

Examples

(i) Write the expanded form of 5,62,57,825

Solution: $5 \times 1,00,00,000 + 6 \times 10,00,000 + 2 \times 1,00,000 + 5 \times 10,000 + 7 \times 1,000 + 8 \times 100 + 2 \times 10 + 5 \times 1$

(ii) Find the sum of place values of all 5s in 5,62,57,825.

Solution: 5 at Crores place: $5 \times 1,00,00,000 = 5,00,00,000$

5 at Ten Thousands: $5 \times 10,000 = 50,000$

5 at Ones place: $5 \times 1 = 5$

Sum = $5,00,00,000 + 50,000 + 5 = 5,00,50,005$

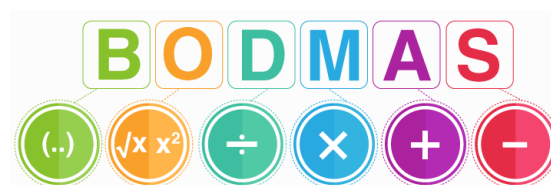
1.5 Rule of BODMAS

Types of Bracket

- Round brackets or parenthesis (), Curly brackets or braces { }, Square brackets [], Bar or Vinculum '—'

Operation 'of'

O in the BODMAS rule stands for 'of' which means multiplication.



Examples

(i) Simplify: (a) $7 - \{13 - 2(4 \text{ of } -4)\}$

Solution: $7 - \{13 - 2(4 \text{ of } -4)\}$
 $= 7 - \{13 - 2(4 \times -4)\}$
 $= 7 - \{13 - 2(-16)\} = 7 - \{13 - (-32)\}$
 $= 7 - \{13 + 32\} = 7 - 45 = -38$

(b) 81 of $[59 - \{7 \times 8 + (13 - 2 \text{ of } 5)\}]$

$= 81 \times [59 - \{7 \times 8 + (13 - 2 \times 5)\}]$
 $= 81 \times [59 - \{7 \times 8 + (13 - 10)\}]$
 $= 81 \times [59 - \{56 + 3\}]$
 $= 81 \times [59 - 59] = 81 \times 0 = 0$



Practice Questions - 1

- (i) Solve the following using the number line concept and check using the standard algorithm.
(a) $33 + 11$ (b) $(+45) - (13)$ (c) $(-67) - (-13)$ (d) $(-38) - (+42)$
- (ii) Simplify $23 - 13 + 23$.
- (iii) Observe the pattern and fill up the boxes.
(a) $6 \times 4 = 24 \therefore 24 \div \square = 6$ (b) $8 \times -5 = -40 \therefore \square \div -5 = 8$
- (iv) A Gardner plans to plant 630 trees in 21 rows each containing the same number of trees, then how many trees will be there in each row?
(A) 300 (B) 30
(C) 21 (D) 3
- (v) Find the greater one: $22 \times -36 + (-42) \div (-3)$ or $9 \times 56 \times (-8)$.
- (vi) Identify the property used in the following.
 $2 \times 13 + 8 \times 13 = (2 + 8) \times 13$
(A) Commutative (B) Closure
(C) Associative (D) Distributive
- (vii) Simplify: $52 - 4$ of $(17 - 12) + 4 \times 7$.

2. PRIME FACTORISATION & DIVISIBILITY RULES

2.1 Factors and Multiples

Before we can understand prime factorisation, we must first understand what factors and multiples are.

2.2 What is a Factor?

A factor of a number is a whole number that divides the given number exactly (with remainder 0).

Another way to think of it: If $a \times b = n$, then both a and b are factors of n .

Examples

(i) Factors of 12

Solution: $1 \times 12 = 12 \rightarrow 1$ and 12 are factors

$2 \times 6 = 12 \rightarrow 2$ and 6 are factors

$3 \times 4 = 12 \rightarrow 3$ and 4 are factors

Therefore, all factors of 12 are: 1, 2, 3, 4, 6, 12

Key Facts About Factors:

- 1 is a factor of every number
- Every number is a factor of itself
- Factors are always less than or equal to the number (except when the number is 0)
- Factors always come in pairs

2.3 What is a Multiple?

A multiple of a number is the product obtained when that number is multiplied by any whole number.

Examples

(i) Multiples of 5

Solution: $5 \times 1 = 5$

$5 \times 2 = 10$

$5 \times 3 = 15$

$5 \times 4 = 20$

Therefore, the first five multiples of 5 are: 5, 10, 15, 20, 25, ...

Key Facts About Multiples:

- Every number is a multiple of itself ($n \times 1 = n$)
- Every number is a multiple of 1 ($1 \times n = n$)
- Multiples of a number are always greater than or equal to that number
- There are infinitely many multiples of any number

Examples

(i) Find all factors of 24.

Solution: Method: Find all pairs whose product is 24

$$1 \times 24 = 24 \rightarrow \text{factors: } 1, 24$$

$$2 \times 12 = 24 \rightarrow \text{factors: } 2, 12$$

$$3 \times 8 = 24 \rightarrow \text{factors: } 3, 8$$

$$4 \times 6 = 24 \rightarrow \text{factors: } 4, 6$$

1, 2, 3, 4, 6, 8, 12, 24

(ii) Write the first 6 multiples of 7.

Solution: $7 \times 1 = 7$, $7 \times 2 = 14$, $7 \times 3 = 21$, $7 \times 4 = 28$, $7 \times 5 = 35$, $7 \times 6 = 42$
7, 14, 21, 28, 35, 42

(iii) Is 45 a multiple of 9? Is 9 a factor of 45?

Solution: $45 \div 9 = 5$ exactly (no remainder)

Yes, 45 is a multiple of 9 (because $9 \times 5 = 45$)

Yes, 9 is a factor of 45 (because it divides 45 exactly)

(iv) Find the common factors of 18 and 24.

Solution: Factors of 18: 1, 2, 3, 6, 9, 18

Factors of 24: 1, 2, 3, 4, 6, 8, 12, 24

Common factors (appear in both lists): 1, 2, 3, 6

2.4 Prime and Composite Numbers

Numbers can be classified based on how many factors they have. This classification is crucial for prime factorisation.

(a) Prime Numbers

A prime number is a natural number greater than 1 that has exactly two factors: 1 and the number itself.

Examples: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, ...

- The only even prime number is 2 (all other even numbers are divisible by 2)
- 1 is NOT a prime number (it has only one factor, not two)
- The smallest prime number is 2
- There are infinitely many prime numbers

(b) Composite Numbers

A composite number is a natural number greater than 1 that has more than two factors.

Examples: 4, 6, 8, 9, 10, 12, 14, 15, 16, 18, 20, 21, 22, 24, 25, ...

- Every composite number can be expressed as a product of prime numbers
- 4 is the smallest composite number
- All even numbers greater than 2 are composite

Note: 1 is neither prime nor composite. It is called a UNIT. It has only one factor (itself).

2.5 Prime Factorisation

Prime factorisation is the process of expressing a composite number as a product of its prime factors. This is one of the most important skills in number theory.

Fundamental Theorem of Arithmetic:

Every composite number can be expressed as a product of prime numbers in exactly ONE way (ignoring the order of factors).

Example: $60 = 2 \times 2 \times 3 \times 5$ (this is the ONLY way to write 60 as a product of primes)

(a) Division Method

This method involves repeatedly dividing the number by the smallest possible prime number until the quotient is 1. Here's the process given below:

- Divide by the smallest prime: Divide the given number by the smallest prime number that divides it evenly.
- Repeat: Divide the resulting quotient by the smallest prime that divides it evenly.
- Continue until 1: Repeat this process until the quotient becomes 1.
- Prime factors: The prime factors are the divisors used in the division process.

2	40
2	20
2	10
5	5
	1

Examples

(i) Find the Prime Factorization of 60 using Division Method.

Solution: Prime Factorization of 60 by Division Method

2	60
2	30
3	15
5	5
	1

Prime Factorization of $60 = 2 \times 2 \times 3 \times 5$

(ii) Find the prime factors of 980 by Division Method and then give the difference between the highest prime factor and smallest prime factor of 980.

Solution:

2	980
2	490
5	245
7	49
7	7
	1

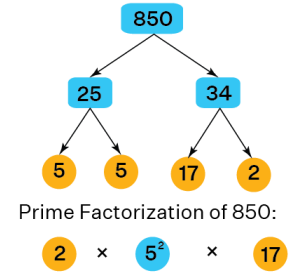
$980 = 2 \times 2 \times 5 \times 7 \times 7$

$\therefore$ The prime factors of 980 are 2, 5 and 7.

The highest prime factor and smallest prime factor of 980 are 7 and 2 respectively the difference $= 7 - 2 = 5$.

(b) Factor Tree Method

Break the number into any two factors, then continue breaking until all branches end in primes.

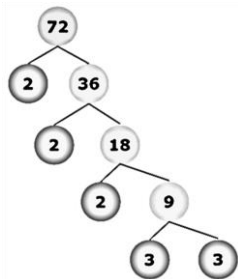


Examples

(i) Write the prime factors of the following numbers using the factor tree method.

(a) 72

Solution: (a)

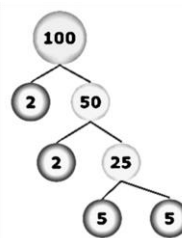


$$72 = 2 \times 2 \times 2 \times 3 \times 3$$

$\therefore$ The prime factors of 72 are 2 and 3.

(b) 100

(b)



$$100 = 2 \times 2 \times 5 \times 5$$

$\therefore$ The prime factors of 100 are 2 and 5.

2.6 Divisibility Rules

Divisibility rules allow us to check if a number is divisible by another number WITHOUT actually performing the division. These are incredibly useful shortcuts.

Divisible by	Rule	Example
2	Last digit is 0, 2, 4, 6, or 8	24, 136, 5678
3	Sum of digits divisible by 3	123 (1+2+3=6)
4	Last 2 digits divisible by 4	1324 (24÷4=6)
5	Last digit is 0 or 5	35, 100, 1255
6	Divisible by both 2 and 3	24, 42, 108
7	(Last digit $\times 2$) subtracted from rest divide by 7	203: 20-6=14
8	Last 3 digits divisible by 8	12,416 (416÷8=52)
9	Sum of digits divisible by 9	459 (4+5+9=18)
10	Last digit is 0	30, 150, 1000
11	Diff of odd/even position sums = 0 or div by 11	1331

Examples

(i) **Check if 1248 is divisible by 3, 4, and 6.**

Solution: Divisible by 3? Sum = $1+2+4+8 = 15$. $15 \div 3 = 5$. Yes, divisible by 3.

Divisible by 4? Last 2 digits = 48. $48 \div 4 = 12$. Yes, divisible by 4.

Divisible by 6? Divisible by both 2 (even) and 3 (proved above). Yes, divisible by 6.

(ii) **Is 5436 divisible by 9?**

Solution: Sum of digits = $5+4+3+6 = 18$

$18 \div 9 = 2$. Yes, 18 is divisible by 9.

Therefore, 5436 is divisible by 9.

(iii) **Check if 7,128 is divisible by 8.**

Solution: Last 3 digits = 128

$128 \div 8 = 16$ exactly

Yes, 7,128 is divisible by 8.

(iv) **Is 10,824 divisible by 11?**

Solution: Odd positions (from right): $4 + 8 + 1 = 13$

Even positions: $2 + 0 = 2$

Difference: $13 - 2 = 11$

11 is divisible by 11, so 10,824 is divisible by 11.

(v) **Is the first number divisible by the second? Use prime factorization.**

(a) 225 and 27

(b) 96 and 24

Solution: (a) The prime factorization of 225 is $3 \times 3 \times 5 \times 5$ and 27 is $3 \times 3 \times 3$

No, 3 occurs twice in the prime factorization of 225 but thrice in the prime factorization of 27.

So, 225 is not divisible by 27.

(b) The prime factorization of 96 is $2 \times 2 \times 2 \times 2 \times 2 \times 3$ and 24 is $2 \times 2 \times 2 \times 3$.

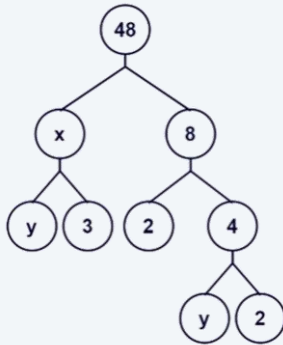
Yes, all the prime factors of 24 occur in the prime factorization of 96.

So, 96 is divisible by 24.



Practice Questions - 2

- (i) Find the value of $x \div y$



- (A) 2 (B) 6 (C) 4 (D) 3
- (ii) What is the smallest number whose prime factorization has three different prime numbers
 (A) 8 (B) 12 (C) 105 (D) 102
- (iii) What least number should be assigned to * so that number $65*347$ is divisible by 11?
 (A) 5 (B) 6 (C) 4 (D) 7
- (iv) Which of the following numbers is divisible by 6?
 (A) 60466176 (B) 59624565
 (C) 45879632 (D) both (A) and (B)
- (v) Find the smallest digit that can replace * to make $34*2$ divisible by 3.
 (A) 2 (B) 4 (C) 1 (D) 0
- (vi) Prime factorization of 54 is:
 (A) 2×27 (B) $2 \times 3 \times 9$ (C) 54×1 (D) 2×3
- (vii) Which of the following number is the product of exactly three distinct prime number?
 (A) 20 (B) 165 (C) 45 (D) 147
- (viii) Sum of the number of primes between 16 to 80 and 90 to 100 is i
 (A) 20 (B) 18 (C) 17 (D) 16
- (ix) Is the first number is divisible by the second? The numbers are 343 and 17.
- (x) The first number has prime factorization $2 \times 3 \times 7$ and the second number has prime factorization $3 \times 7 \times 11$. Are they co-prime?

3. EXPONENTS AND POWER

3.1 Laws of Exponents

Law 1: Negative exponents

If a and m are two non-zero integers, then

$$a^{-m} = \frac{1}{a^m}$$

For Example: $5^{-1} = \frac{1}{5}$

Law 2: Multiplication of exponential forms with the same base

If a is any non-zero integer and m and n are integers, then

$$a^m \times a^n = a^{m+n}$$

For Example: $2^3 \times 2^5 = 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 = 2^8 = 2^{3+5}$

Law 3: Division of exponential forms with the same base

If a is a non-zero integer and m and n are two real numbers such that $m > n$, then

$$a^m \div a^n = a^{m-n}$$

For Example: $3^5 \div 3^3 = \frac{3 \times 3 \times 3 \times 3 \times 3}{3 \times 3 \times 3} = 3 \times 3 = 3^2 = 3^{5-3}$

Law 4: Multiplication of exponential forms with the same exponents and different bases

If a and b are non-zero integers and m is a real number, then

$$a^m \times b^m = (a \times b)^m = (ab)^m$$

For Example:

$$2^4 \times 3^4 = 2 \times 2 \times 2 \times 2 \times 3 \times 3 \times 3 \times 3 = 6 \times 6 \times 6 \times 6 = 6^4 = (2 \times 3)^4$$

Law 5: Division of exponential forms with the same exponents and different bases

If a and b are any two non-zero integers that have the same exponent m , then $a^m \div b^m = (a \div b)^m$

$$\frac{a^m}{b^m} = \left(\frac{a}{b}\right)^m$$

For Example: $5^3 \div 2^3 = \frac{5 \times 5 \times 5}{2 \times 2 \times 2} = \frac{5}{2} \times \frac{5}{2} \times \frac{5}{2} = \left(\frac{5}{2}\right)^3$

Law 6: Power of a Power

If ' a ' is any non-zero integer and m and n be any two integers then

$$[(a)^m]^n = a^{mn}$$

For Example: $(2^2)^3 = 2^2 \times 2^2 \times 2^2 = 2^{2+2+2} = 2^6 = 2^{2 \times 3}$

Law 7: Zero Exponent

In general, $a^m \div a^m = a^{m-m} = a^0$ and also

$$a^0 = 1$$

$$\frac{a^m}{a^m} = \frac{a \times a \times a \times a \times a \times a \times \dots, m \text{ times}}{a \times a \times a \times a \times a \times a \times \dots, m \text{ times}} = 1$$

Examples

- (i) Express $\left(-\frac{3}{2}\right)^{-3}$ as a base having a positive exponent.

Solution: $\left(-\frac{3}{2}\right)^{-3} = \left(-\frac{2}{3}\right)^3$

- (ii) Evaluate: $\left(-\frac{2}{7}\right)^{-4} \times \left(-\frac{2}{7}\right)^2$

Solution: In both powers, bases are the same so,
We can use $a^m \times a^n = a^{m+n}$

$$\left(-\frac{2}{7}\right)^{-4} \times \left(-\frac{2}{7}\right)^2 = \left(-\frac{2}{7}\right)^{-4+2} = \left(-\frac{2}{7}\right)^{-2}$$

- (iii) Find the value of n , when $\left(\frac{5}{3}\right)^{2n+1} \times \left(\frac{5}{3}\right)^5 = \left(\frac{5}{3}\right)^{n+2}$

Solution: $\left(\frac{5}{3}\right)^{2n+1+5} = \left(\frac{5}{3}\right)^{n+2}$

$$\Rightarrow \left(\frac{5}{3}\right)^{2n+6} = \left(\frac{5}{3}\right)^{n+2}$$

Since the bases are the same and by equating the powers, we get $2n + 6 = n + 2$

$$2n - n = 2 - 6 \Rightarrow n = -4$$

- (iv) If $a = \left(\frac{2}{5}\right)^2 \div \left(\frac{9}{5}\right)^0$ find the value of a^{-3} .

Solution: $a^{-3} = \left[\left(\frac{2}{5}\right)^2 \div \left(\frac{9}{5}\right)^0\right]^{-3} = \left[\left(\frac{2}{5}\right)^2 \div 1\right]^{-3} = \left[\left(\frac{2}{5}\right)^2\right]^{-3} = \left(\frac{2}{5}\right)^{-6}$

3.2 Use of Exponents in Expressing Large Numbers (Standard Form)

The number which preceded the power of 10 should be less than 10. Such a notation is called **standard** or **scientific notation**.

In general: $K \times 10^n$, where $1 < K < 10$, $n \in \mathbb{I}$

Examples

- (i) Write the following numbers in standard form.

(a) 43,40,000 (b) 1,73,000 (c) 1,40,000 (d) 602000000000000

Solution: (a) It is clear that $43,40,000 = 434 \times 10000 = 4.34 \times 10^6$ (b) $1,73,000 = 1.73 \times 10^5$

(c) $1,40,000 = 1.4 \times 10^5$

(d) $602000000000000 = 6.02 \times 10^{14}$

(ii) Express the following numbers in the usual form.

(a) 7.51×10^5

(b) 1.001×10^9

(c) 6.678×10^8

Solution: We have,

(a) $7.51 \times 10^5 = 7.51 \times 100000 = 751000$

(b) $1.001 \times 10^9 = 1,00,10,00,000$

(c) $6.678 \times 10^8 = 66,78,00,000$

3.3 Unit Conversions — Length and Weight

In the metric system, we use Powers of 10 to move between units. Instead of writing a dozen zeros, we use exponents to keep things clean.

Unit	Scientific Notation (Meters)	Power
Kilometer (km)	10^3 m	1,000 m
Meter (m)	10^0 m	1m
Centimeter (cm)	10^{-2} m	0.01 m
Millimeter (mm)	10^{-3} m	0.001 m

The Rule: When you convert "up" (e.g., cm to m), you subtract from the exponent. When you move "down" (e.g., km to m), you add to it.

Example: If a mountain height is 5.5 km , and you want it in millimeters:

$$5.5 \times 10^3 \text{ (to meters) } \times 10^3 \text{ (to mm) } = 5.5 \times 10^6 \text{ mm}$$

(a) Length Conversions

To Convert	Multiply By	Example
km to m	$\times 1000$	$3.5 \text{ km} = 3.5 \times 1000 = 3500 \text{ m}$
m to cm	$\times 100$	$4.2 \text{ m} = 4.2 \times 100 = 420 \text{ cm}$
cm to mm	$\times 10$	$6.5 \text{ cm} = 6.5 \times 10 = 65 \text{ mm}$
m to km	$/ 1000$	$2500 \text{ m} = 2500/1000 = 2.5 \text{ km}$
cm to m	$/ 100$	$350 \text{ cm} = 350/100 = 3.5 \text{ m}$
mm to cm	$/ 10$	$85 \text{ mm} = 85/10 = 8.5 \text{ cm}$

(b) Weight Conversions

To Convert	Multiply By	Example
kg to g	$\times 1000$	$2.5 \text{ kg} = 2.5 \times 1000 = 2500 \text{ g}$
g to mg	$\times 1000$	$3.2 \text{ g} = 3.2 \times 1000 = 3200 \text{ mg}$
g to kg	$/ 1000$	$4500 \text{ g} = 4500/1000 = 4.5 \text{ kg}$
mg to g	$/ 1000$	$6500 \text{ mg} = 6500/1000 = 6.5 \text{ g}$

Examples

(i) The distance between Rita's and Vinay's house is 53,825 m and between Vinay's and Tarun's is 25,783 m. Find the total distance Rita covers (in km) going to Vinay's then Tarun's.

Solution: Total distance = 53,825 m + 25,783 m = 79,608 m

Convert to km: $79,608 \text{ m} / 1000 = 79.608 \text{ km}$ (or 79 km 608 m)

(ii) A box has 40,000 tablets each weighing 10 mg. Find total weight in grams and kilograms.

Solution: Total weight = $40,000 \times 10 = 4,00,000$ mg

Convert to g: $4,00,000 / 1000 = 400$ g

Convert to kg: $400 / 1000 = 0.4$ kg

(iii) 4 kg 500 g of sand is used from a sack weighing 10 kg. How much sand is left in the sack?

Solution:

$$\begin{array}{r}
 \text{kg} \qquad \qquad \text{g} \\
 \textcircled{9} \quad \textcircled{0} \textcircled{10} \\
 10 \quad 1000 \\
 - \quad 4 \quad 500 \\
 \hline
 5 \quad 500 = 5 \text{ kg } 500 \text{ g}
 \end{array}$$

Therefore, 5 kg 500 g sand is left in the sack.



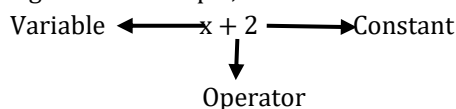
Practice Questions - 3

- (i) $2^2 \times 2^3 \times 2^4$ is equal to
 (A) 2^{24} (B) 2^5 (C) 2^9 (D) 2^{-5}
- (ii) $\frac{5^4}{5^2}$ is equal to
 (A) 5^{-5} (B) 5^6 (C) 5^2 (D) 5^{-2}
- (iii) $100^0 + 20^0 + 5^0$ is equal to
 (A) 125 (B) 25 (C) 3 (D) $\frac{1}{125}$
- (iv) Find the value of $3^4 \left[\left(\frac{2}{3}\right)^2 + \frac{2}{3} - \left(\frac{2}{3}\right)^3 \right]$.
 (A) 86 (B) 66 (C) 68 (D) 88
- (v) Value of $\left(\frac{1024}{243}\right)^{\frac{3}{5}}$ is
 (A) $\frac{128}{27}$ (B) $\frac{32}{27}$ (C) $\frac{64}{27}$ (D) $\frac{32}{9}$
- (vi) 700,000,000 is equal to
 (A) 7×10^8 (B) 7×10^6 (C) 7×10^7 (D) 7×10^9
- (vii) 0.09×10^{10} is equal to
 (A) 900,000,000 (B) 9,000,000 (C) 9000 (D) 1
- (viii) Write the following number in the usual form.
 (a) 3.49×10^4 (b) 1.11×10^6
- (ix) Express the following number in the form $\mathbf{K} \times 10^n$, where $\mathbf{K}$ is a number and $\mathbf{n}$ is an integer:
 4,176,300,000
- (x) With a global human population of about 8×10^9 and about 4×10^5 African elephants, can we say that there are nearly 20,000 people for every African elephant?
- (xi) A man walks from A to B (9 km 500 m), then B to C (1,500 m), then C to D (20 km). Find total distance in km.
- (xii) Convert: (a) 4.75 km to metres (b) 83,500 m to km (c) 2.3 kg to grams
- (xiii) If a sugar sachet weighs 5 g, how much will it be in milligrams?
- (xiv) A restaurant owner uses 5 kg 200 g, 8 kg 900 g, and 12 kg 600 g of onions over 3 days. What is the total weight of onions used by the restaurant owner in 3 days?
- (xv) Aarav is helping his grandfather at the fruit stall. He lifts two baskets of apples weighing 2 kg 100 g and 3 kg 950 g. What is the total weight of apples he lifted?

4. ALGEBRAIC EXPRESSIONS

4.1 Introduction

An algebraic expression is defined as “the combination of variables, constants, and operators like ‘+’, ‘-’, ‘×’ & ‘÷’ signs”. For example,



(i) Term: The various parts of an algebraic expression connected by + or – sign are called terms of the expression.

For Example: $2x + 4y^2 - 3z$ has three terms, namely, $+2x$, $+4y^2$ and $-3z$.

$2xy - 4y^2$ has two terms namely $+2xy$ and $-4y^2$.

(ii) Like terms and Unlike terms: The terms having the same literal factors or coefficient are called like terms and those having different literal factors or coefficient are called unlike terms.

For Example: $-6xy$ and $4yx$ are like terms whereas $2xy$, $-3x^2$, $4xy^2$ are unlike terms.

4.2 Types of Algebraic Expressions

An algebraic expression is classified as a monomial, binomial, or trinomial based on the number of terms that it contains.

(i) Monomial: An algebraic expression that contains only one term is called a monomial.

For Example: $2x$, 3 , $4t$, $9pq$, $2c^2$ etc.

(ii) Binomial: An algebraic expression that contains two unlike terms is called a binomial.

For Example: $6x + 2x^2$ is binomial since it contains two unlike terms namely, $6x$ and $2x^2$.

(iii) Trinomial: An algebraic expression that contains three unlike terms is called a trinomial.

For Example: $x + x^2 - 3x^3$ is trinomial since it contains three unlike terms namely, x , x^2 and $-3x^3$.

4.3 Operations on Algebraic Expressions

The sum or difference of several like terms is another like terms whose coefficient is the sum or difference of those like terms.

In addition, or subtraction, we can add or subtract only like terms together means those terms having the same algebraic factors.

Addition of Algebraic Expressions

(a) Horizontal Method

Consider that we have to add three algebraic expressions $5xy - 3x^2 - 12y + 5x$, $xy - 3x - 12yz + 5x^3$ and $y - 6x^2 - zy + 5x^3$

Step 1: Write the given algebraic expressions using the symbol of addition.

$$(5xy - 3x^2 - 12y + 5x) + (xy - 3x - 12yz + 5x^3) + (y - 6x^2 - zy + 5x^3)$$

Step 2: Open the brackets and multiply the signs.

$$5xy - 3x^2 - 12y + 5x + xy - 3x - 12yz + 5x^3 + y - 6x^2 - zy + 5x^3$$

Step 3: Now, combine the like terms.

$$(5xy + xy) + (-3x^2 - 6x^2) + (-12y + y) + (5x - 3x) + (-12yz - zy) + (5x^3 + 5x^3)$$

Step 4: Add the coefficients. Keep the variables and exponents on the variables the same.

$$6xy - 9x^2 - 11y + 2x - 13yz + 10x^3$$

(b) Column Method

In this method, the given expression must be written column-wise one below the other. For adding two or more algebraic expressions the like terms of both the expressions are grouped. The coefficients of like terms are added together using simple addition techniques and the variable which is common is retained as it is.

+5xy	-3x ²	-12y	+5x	+0yz	+0x ³
+xy	+0x ²	+0y	-3x	-12yz	+5x ³
+0xy	-6x ²	+1y	+0x	-yz	+5x ³
6xy	-9x ²	-11y	+2x	-13yz	+10x ³

Subtraction of Algebraic Expressions

To subtract an algebraic expression from another, we change the sign (from '+' to '-' or from '-' to '+') of all the terms of the expression which is to be subtracted, and then the two expressions are added.

Subtract $x^2y - 2x^2 - zy + 5$ from $y^3 + 3x^2y - 6x^2 - 6zy + 7x^3$

Solution: Let's solve this problem in two ways.

(a) Horizontal Method

Step 1: Write the given algebraic expressions using the subtraction symbol.

$$(y^3 + 3x^2y - 6x^2 - 6zy + 7x^3) - (x^2y - 2x^2 - zy + 5)$$

Step 2: Open the brackets and multiply the signs.

$$y^3 + 3x^2y - 6x^2 - 6zy + 7x^3 - x^2y + 2x^2 + zy - 5$$

Step 3: Now, combine the like terms.

$$y^3 + (3x^2y - x^2y) + (-6x^2 + 2x^2) + (-6zy + zy) + (7x^3) + (-5)$$

Step 4: Add the coefficients. Keep the variables and exponents on the variables the same.

$$y^3 + 2x^2y - 4x^2 - 5zy + 7x^3 - 5$$

(b) Column Method

For subtracting two or more algebraic expressions, it's a better practice to write the expressions to be subtracted below the expression from which it is to be subtracted. Like terms are placed below each other. The sign of each term that is to be subtracted is reversed and then the resulting expression is added normally.

y ³	+3x ² y	-6x ²	-6zy	+7x ³	+0
+0y ³	+x ² y	-2x ²	-zy	+0x ³	+5
-	-	+	+	-	-
y ³	+2x ² y	-4x ²	-5zy	+7x ³	-5

Here the result in standard form is $y^3 + 7x^3 + 2x^2y - 4x^2 - 5zy - 5$.

Examples

(i) **Add the following: $3xy$, $10xy$ and $5xy$.**

Solution: The sum of the numerical coefficients of the given like terms is $3 + 10 + 5 = 18$.
Hence, $3xy + 10xy + 5xy = 18xy$.

(ii) **Add $4x^2 - 3x + 1$ to the sum of $3x^2 - 2x$ and $x + 7$**

Solution: Sum of $3x^2 - 2x$ and $x + 7$
 $= (3x^2 - 2x) + (x + 7)$
 $= 3x^2 - 2x + x + 7 = 3x^2 - x + 7$
 Now, the required expression
 $= 4x^2 - 3x + 1 + 3x^2 - x + 7$
 $= 4x^2 + 3x^2 - 3x - x + 1 + 7$
 $= 7x^2 - 4x + 8$

(iii) **Add the following: $3x + 4y + 5z$ and $2x - 3y - 4z$.**

Solution: Horizontal Method,
 $= (3x + 4y + 5z) + (2x - 3y - 4z)$
 $= (3x + 2x) + (4y - 3y) + (5z - 4z)$
 $= (3 + 2)x + (4 - 3)y + (5 - 4)z = 5x + y + z$
 Column Method

$$\begin{array}{r} 3x + 4y + 5z \\ + 2x - 3y - 4z \\ \hline 5x + y + z \end{array}$$

Multiplication of Algebraic Expression

The following are the rules of signs and the laws of exponents that are used in the multiplication of algebraic expressions.

Rule 1: (i) Multiply numerical coefficients, (ii) Multiply literal coefficients, (iii) Multiply results.

For Example:

(i) $3x \times 5y = (3 \times 5) \times (x \times y) = 15 \times xy = 15xy$.

Rule 2: Signs in multiplication:

The product of two factors with like signs is positive, and the product of two factors with unlike signs is negative. Thus if a and b are two positive numbers then,

- (i) $(+a) \times (+b) = +(ab)$; e.g. $2 \times 3 = 6$
 (ii) $(+a) \times (-b) = -(ab)$; e.g. $\frac{2}{3} \times \frac{-5}{7} = \frac{-10}{21}$
 (iii) $(-a) \times (+b) = -(ab)$; e.g. $\frac{-2}{3} \times \frac{5}{7} = \frac{-10}{21}$
 (iv) $(-a) \times (-b) = +(ab)$; e.g. $\frac{-2}{3} \times \frac{-5}{7} = \frac{10}{21}$

Rule 3: Laws of exponents in multiplication:

To multiply the powers of the same base, keep the base same and add the exponents.

$$\begin{array}{ll} \text{Thus, } a \cdot a^2 = a^{1+2} = a^3, & x^3 \cdot x^5 \cdot x = x^{(3+5+1)} = x^9. \\ a^3 b^4 \cdot a^2 b^7 = a^5 b^{11}, & 3^2 \cdot 3^3 \cdot x \cdot x^4 \cdot x^5 \cdot y^2 \cdot y = 3^5 x^{10} y^3. \end{array}$$

Examples

(i) $4a^2b \times (-6a^3b^2c)$ (b) $-\frac{6}{8}x^4yz \times 24x^2y^2z^3$.

Solution: (a) $4a^2b \times -6a^3b^2c = (4 \times -6) \times (a^2b \times a^3b^2c) = -24a^5b^3c$.

(b) $-\frac{6}{8}x^4yz \times 24x^2y^2z^3 = \left(-\frac{6}{8} \times 24\right) \times (x^4yz \times x^2y^2z^3) = -18x^6y^3z^4$.

(a) Multiplication of a Monomial by a Monomial

Examples

(i) (a) $3x^3y^2 \times 4x^2y$ (b) $\frac{4}{5}ab \times \frac{-3}{2}a^2bc$ (c) $\frac{-2}{3}x \times \frac{3}{4}y \times \frac{-4}{5}z$

Solution: (a) $3x^3y^2 \times 4x^2y = (3 \times 4)x^{3+2}y^{2+1} = 12x^5y^3$

(b) $\frac{4}{5}ab \times \frac{-3}{2}a^2bc = \left(\frac{4}{5} \times \frac{-3}{2}\right)a^{1+2}b^{1+1}c^1 = \frac{-6}{5}a^3b^2c$

(c) $\frac{-2}{3}x \times \frac{3}{4}y \times \frac{-4}{5}z = \left(\frac{-2}{3} \times \frac{3}{4} \times \frac{-4}{5}\right)x \times y \times z = \frac{2}{5}xyz$

(b) Multiplication of a Binomial by a Monomial

To multiply a binomial by a monomial, use the rule of distributive property:

$$a \times (b + c) = (a \times b) + (a \times c)$$

Examples

(i) **Multiply:** (a) $3x^2 + 4xy$ by $2x$ (b) $-4a(a + 3b)$ (c) $-4a(a - 3b)$

Solution: (a) $2x[3x^2 + 4xy] = (2x \times 3x^2) + (2x \times 4xy) = 6x^3 + 8x^2y$

(b) $-4a[a + 3b] = (-4a \times a) + (-4a \times 3b) = -4a^2 - 12ab$

(c) $-4a[a - 3b] = (-4a \times a) + (-4a \times (-3b)) = -4a^2 + 12ab$.

(c) Multiplication of a Binomial by a Binomial

Suppose we have to multiply two binomials $(a + b)$ and $(c + d)$. We can do so by using the distributive law of multiplication over addition, twice as shown below.

$$\begin{aligned} (a + b) \times (c + d) &= a \times (c + d) + b \times (c + d) && \text{(Applying distributive law)} \\ &= (a \times c + a \times d) + (b \times c + b \times d) && \text{(Applying distributive law again)} \\ &= ac + ad + bc + bd \end{aligned}$$

Thus, to multiply any two binomials, we multiply each term of one binomial by each term of the other and then add the products.

Examples

(i) **Multiply $3a + 2b$ and $5a + 7b$.**

$$\begin{aligned}\text{Solution: } (3a + 2b)(5a + 7b) &= 3a(5a + 7b) + 2b(5a + 7b) \\ &= 15a^2 + 21ab + 10ab + 14b^2 \\ &= 15a^2 + 31ab + 14b^2\end{aligned}$$

(ii) **Simplify: $(5x - 3y)(-x + 7y) - (2x + 5y)(3x - 4y)$.**

$$\begin{aligned}\text{Solution: } &\text{We have,} \\ &(5x - 3y)(-x + 7y) - (2x + 5y)(3x - 4y) \\ &= 5x(-x + 7y) - 3y(-x + 7y) - [2x(3x - 4y) + 5y(3x - 4y)] \\ &= (-5x^2 + 35xy + 3xy - 21y^2) - (6x^2 - 8xy + 15xy - 20y^2) \\ &= (-5x^2 + 38xy - 21y^2) - (6x^2 + 7xy - 20y^2) \\ &= -5x^2 + 38xy - 21y^2 - 6x^2 - 7xy + 20y^2 \\ &= (-5x^2 - 6x^2) + (38xy - 7xy) + (-21y^2 + 20y^2) = -11x^2 + 31xy - y^2.\end{aligned}$$

4.4 Algebraic Identities

Identity 1. $(a + b)^2 = a^2 + 2ab + b^2$

Identity 2. $(a - b)^2 = a^2 - 2ab + b^2$

Identity 3. $(a + b)(a - b) = a^2 - b^2$

Identity 4. $(x + a)(x + b) = x^2 + (a + b)x + ab$

Examples

(i) **Find the following products using Identity:**

(a) $(x + 3)(x - 3)$

(b) $(y + 2)^2$

(c) $(x + 5)(x - 3)$

Solution: **(a)** Let us compare, $(x + 3)(x - 3)$ with $(a + b)(a - b)$. We observe that;

$$x = a, b = 3$$

Therefore, using Identity 3, we have

$$\begin{aligned}(x + 3)(x - 3) &= x^2 - 3^2 \\ &= x^2 - 9\end{aligned}$$

(b) $(y + 2)^2 = y^2 + 2 \cdot y \cdot 2 + 2^2 = y^2 + 4y + 4$

(c) Let us compare, $(x + 5)(x - 3)$ with $(x + a)(x + b)$. We observe that;

$$a = 5, b = -3$$

Therefore, using Identity 4, we have;

$$\begin{aligned}(x + 5)(x - 3) &= x^2 + \{5 + (-3)\}x + 5 \times (-3) \\ &= x^2 + 2x - 15\end{aligned}$$



Practice Questions - 4

- (i) Subtract $3x^2 - 6x - 4$ from $5 + x - 2x^2$.
- (ii) What is the coefficient of x^2 when you add $2x^4 + 5x^3$ and $6x^4 - 7x^3 + 5$?
- (iii) What expression must be added to $1 - x + x^2 - 2x^3$ to get x^3 ?
- (iv) Subtract $4pq - 5q^2 - 3p^2$ from $5p^2 + 3q^2 - pq$.
 (A) $8p^2 + 8q^2 - 5pq$ (B) $8p^2 + 8q^2 + 5pq$
 (C) $8p^2 - 8q^2 - 5pq$ (D) $8p^2 - 8q^2 + 5pq$
- (v) The value of $(x + y)(x - y) + (y - z)(y + z) + (z - x)(z + x)$ is equal to
 (A) $3x^2$ (B) $3y^2$
 (C) $3z^2$ (D) 0
- (vi) Which of the following is the value of $\left(x + \frac{1}{x}\right)^2$?
 (A) $x^2 + \frac{1}{x^2}$ (B) $x^2 - \frac{1}{x^2}$
 (C) $x^2 + \frac{1}{x^2} + 2$ (D) $x^2 + \frac{1}{x^2} + 2x$
- (vii) Simplify the following using the formula: $(a + b)(a - b) = a^2 - b^2$
 $(82)^2 - (18)^2$
- (viii) Find the product of the following binomials
 (a) $(2x + y)(2x + y)$ (b) $(a + 2b)(a - 2b)$
- (ix) Using identity evaluate the following number
 (a) $(999)^2$ (b) $(703)^2$
- (x) Which is larger? Find without fully computing the product.
 (a) 14×26 or 16×24
 (b) 25×75 or 26×74

5. SIMPLE EQUATIONS

5.1 Equation

(a) What is an Equation?

An equation is a mathematical statement that shows two expressions are equal, connected by an equal sign (=).

Every equation has two sides:

- Left Hand Side (LHS) — the expression on the left of the = sign
- Right Hand Side (RHS) — the expression on the right of the = sign
- An equation is true only when LHS = RHS

Example: $3x + 5 = 20$ Here LHS = $3x + 5$ and RHS = 20

(b) Basic Terms

Term	Meaning	Example
Variable	A letter that represents an unknown number	x, y, z, a, b
Constant	A fixed number that does not change	3, -7, 100, 0
Expression	Variables and constants combined using operations	$3x + 5$, $2y - 7$
Equation	Two expressions connected by an = sign	$3x + 5 = 20$
Linear Equation	the highest index of the unknown's or variables present is one	$2x + 3 = 11$ and $3x + 4y = 8$
Simple Equation	Linear equation with only one variable	$4x - 1 = 15$ and $2x - 5 = x + 3$

5.2 Forming Equations from Statements

Real-life situations can be written as equations by assigning a variable to the unknown quantity.

Statement	Equation
A number x increased by 7 is 15	$x + 7 = 15$
4 times a number y is 28	$4y = 28$
A number z divided by 5 gives 6	$z / 5 = 6$
9 exceeds a number x by 4	$9 = x + 4$
Twice a number minus 3 equals 11	$2x - 3 = 11$
Sum of a number and its double is 18	$x + 2x = 18$

5.3 Solution of an Equation

A value of variable that makes LHS = RHS of an equation and this value of the variable is said to satisfy the equation and is called a solution or root of the equation. Solving an equation means finding the roots of the equation.

Example:

Is $x = 4$ a solution of the equation $3x - 5 = 7$?

LHS = $3(4) - 5 = 12 - 5 = 7 = \text{RHS}$ ✓ So $x = 4$ IS a solution.

Is $x = 3$ a solution of the equation $2x + 1 = 8$?

LHS = $2(3) + 1 = 7 \neq 8 = \text{RHS}$ ✗ So $x = 3$ is NOT a solution.

5.4 Properties of Equality

(a) Addition property: If the same number is added to both the sides of an equality, the equality remains the same. If $x = y$, then $x + z = y + z$.

(b) Subtraction property: If the same number is subtracted from both sides of an equality, the equality remains the same. If $x = y$, then $x - z = y - z$.

(c) Multiplication property: If both sides of an equality are multiplied by the same number, then the equality remains the same. If $x = y$, then $(x)(z) = (y)(z)$.

(d) Division Property: If both sides of an equality are divided by the same non-zero number, then the equality remains the same. If $x = y$, then, $\frac{x}{z} = \frac{y}{z}$, where $z \neq 0$.

5.5 Solving A Simple Equation

(a) Transposition method

In this method, any term of an equation can be taken to the other side with its sign changed without affecting the equality.

Some Rules for Transposition:

(i) While shifting terms from LHS to RHS or RHS to LHS then, their sign changes to from $(-ve)$ to $(+ve)$ or $(+ve)$ to $(-ve)$ respectively.

(ii) While moving the terms from LHS to RHS, which are in multiplication, changes to division and which are in division changes to multiplication.

Examples

(i) Solve the equation $2x + 9 = 19$

Solution: The given equation is $2x + 9 = 19$

Transpose 9 to the RHS

$$\Rightarrow 2x = 19 - 9$$

$$\Rightarrow 2x = 10$$

Now, divide both sides by 2

$$\Rightarrow \frac{2x}{2} = \frac{10}{2} \Rightarrow x = 5$$

Hence, the solution of equation $2x + 9 = 19$ is $x = 5$

(ii) Find the value of x : $\frac{3x}{2} + \frac{x}{3} = x + \frac{x}{6} + 6$

Solution: $\frac{3x}{2} + \frac{x}{3} = x + \frac{x}{6} + 6$

$$\Rightarrow \frac{3x}{2} + \frac{x}{3} - x - \frac{x}{6} = 6$$

$$\Rightarrow \frac{(3 \times 3x) + (2 \times x) - (6 \times x) - (1 \times x)}{6} = 6$$

$$\Rightarrow \frac{9x + 2x - 6x - 1x}{6} = 6$$

$$\Rightarrow 11x - 7x = 6 \times 6$$

$$\Rightarrow 4x = 36 \Rightarrow x = \frac{36}{4} = 9$$

5.6 Applications of Simple Equations

The applications of linear equations are vast and are applicable in numerous real-life situations. To handle real-life situations using algebra, we change the given situation into mathematical statements.

- Convert the real problem into a mathematical statement and frame it in the form of an algebraic expression that clearly defines the problem situation.
- Identify the unknowns in the situation and assign variables to these unknown quantities.
- Read the situation clearly a number of times and cite the data, phrases, and keywords. Sequentially organize the obtained information.
- Write an equation using the algebraic expression and the provided data in the statement and solve it using systematic equation solving techniques.
- Reframe the answer to the problem and consider whether it truly solves the issue. These steps make it simple to solve word problem applications.

Examples

- (i) **If 7 is subtracted from three times a number, the result is 14. Find the number.**

Solution: Let the required number be x . Then,

$$3x - 7 = 14$$

$$3x = 14 + 7$$

$$3x = 21$$

$$x = 7$$

Hence, the required number is 7.

- (ii) **Radha's father is thrice as old as Radha. After 12 years, his age will be twice that of his daughter. Find their present ages.**

Solution: Let Radha's present age be x years.

Then, her father's present age = $3x$ years

Radha's age after 12 years = $(x + 12)$ years

Radha's father's age after 12 years = $(3x + 12)$ years

$$\therefore 3x + 12 = 2(x + 12)$$

$$\Rightarrow 3x + 12 = 2x + 24$$

$$\Rightarrow 3x - 2x = 24 - 12 \quad [\text{transposing } 2x \text{ to LHS and } 12 \text{ to RHS}]$$

$$\Rightarrow x = 12$$

$\therefore$ Radha's present age = 12 years and her father's present age = (3×12) years = 36 years.

- (iii) **The cost of three chairs and four tables is ₹ 2800. If the cost of each chair is ₹ 600, then find the cost of each table (in ₹).**

Solution: The cost of each chair is ₹ 600.

The cost of three chairs = $3 \times 600 = ₹1800$

$\Rightarrow$ Let the cost of each table = ₹ T

$$₹1800 + 4T = ₹2800 \text{ (given)} \Rightarrow 4T = ₹1000 \therefore T = ₹250$$



Practice Questions - 5

- (i) Solve the equation $x - 7 = 8$.
 (A) 25 (B) 20
 (C) 35 (D) 15
- (ii) Solve: $3(x + 3) - 2(x - 1) = 5(x - 5)$
 (A) 9 (B) 5
 (C) 11 (D) 7
- (iii) Solve the equation $5 + \frac{x}{2} = 3x$.
 (A) 2 (B) 5
 (C) -2 (D) None of these
- (iv) Sixteen years hence, a man's age will be 9 times his age 16 years ago. Find his age 5 years hence.
 (A) 25 (B) 20
 (C) 35 (D) 15
- (v) If 5 is subtracted from three times a number, the result is 16. Find the number.
 (A) 9 (B) 5
 (C) 11 (D) 7
- (vi) There were only two candidates who participated in an election. One contestant got 62% votes and was elected by a margin of 144 votes. Find the total number of votes.
- (vii) The ages of A and B are in the ratio 5 : 3. After 6 years, their ages will be in the ratio 7 : 5. Find the sum of their present ages.
- (viii) 184 is divided into two parts such that one-third of one part exceeds one-seventh of the other part by 8, then find the greater part.
- (ix) Solve: $(m - 3)/2 + (m - 2)/3 = -13$
- (x) Radha's father is thrice as old as Radha. After 12 years his age will be twice Radha's age. Find their present ages.

6. RATIO, PROPORTION AND VARIATION

6.1 Ratio

A ratio is the comparison of two quantities of the same kind by division. It shows how many times one quantity is of another.

Notation: The ratio of a to b is written as $a : b$ (read as "a is to b") = a/b , where $b \neq 0$

Properties of a Ratio

- A ratio has no unit — both quantities must be of the same kind
- The consequent (second term) can never be zero ($b \neq 0$)
- Ratio of unlike quantities is not defined (e.g. 5 kg : 15 m is meaningless)
- A ratio remains unchanged when both terms are multiplied or divided by the same non-zero number
- Every ratio must be expressed in its simplest (lowest) form

6.2 Simplest Form of a Ratio

The ratio of two or more quantities is said to be in the simplest form

- If the highest common factor (HCF) of the quantities is 1.
- If the HCF of the quantities is not 1, then each quantity of the ratio is divided by the HCF to convert the ratio into its simplest form.

Examples

(i) Express 24 : 36 in its simplest form.

Solution: The HCF of 24 and 36 is 12. We divide each term by 12.

$$\text{Then, } 24 : 36 = \frac{24}{12} : \frac{36}{12} = 2 : 3$$

$\therefore$ 24 : 36 in its simplest form is 2 : 3.

(ii) If the height of Alia and Ranbir is 120 cm and 160 cm respectively and the ratio of Alia and Deepika's height is 6 : 7. Find the height of Deepika and the ratio of their heights.

Solution: Alia : Deepika = 6 : 7

$$\frac{120}{\text{Deepika}} = \frac{6}{7}$$

$$\Rightarrow \text{Deepika} = 140 \text{ cm}$$

$$\text{Alia : Deepika : Ranbir} = 120 : 140 : 160$$

$$\text{Alia : Deepika : Ranbir} = 6 : 7 : 8$$

6.3 Dividing Some Quantity in the Given Ratio

When we divide a quantity 'p' in the ratio $m : n$, then two parts are $\frac{m}{m+n} \times p$ and $\frac{n}{m+n} \times p$

Examples

- (i) **In a class of 60 pupils, the ratio of the number of boys to the number of girls is 7 : 8. How many boys and girls are there?**

Solution: Given that 7 are boys and 8 are girls so they are 15 together.

Therefore, boys are 7 out of 15, i.e., $\frac{7}{15}$ of 60, and girls are 8 out of 15, i.e. $\frac{8}{15}$ of 60.

$\therefore$ The number of boys = $\frac{7}{15}$ of 60

$$= \frac{7}{15} \times 60 = 7 \times 4 = 28$$

and the number of girls = $\frac{8}{15} \times 60 = 8 \times 4 = 32$

Check: $28 + 32 = 60$

6.4 Proportion

(a) What is Proportion?

The equality of two ratios is called proportion. If $a : b = c : d$, then a, b, c, d are said to be in proportion.

Written as: $a : b :: c : d$ (read as "a is to b as c is to d")

Golden Rule of Proportion:

Product of Extremes = Product of Means

i.e., $a \times d = b \times c$

Examples

- (i) **Check whether the following statements are true or false.**

(a) $12 : 18 = 28 : 56$

(b) 25 people : 130 people = 15 kg : 78 kg

Solution: (a) $12 : 18 = 28 : 56$

The given statement is false.

$$12 : 18 = \frac{12}{18} = \frac{2}{3} = 2 : 3$$

$$28 : 56 = \frac{28}{56} = \frac{1}{2} = 1 : 2$$

They are unequal.

(b) 25 people : 130 people = 15 kg : 78 kg

The given statement is true.

$$25 \text{ people} : 130 \text{ people} = 5 : 26$$

$$15 \text{ kg} : 78 \text{ kg} = 5 : 26$$

They are equal.

(ii) Check whether 63, 42, 33 and 22 are in proportion?

Solution: Let $a = 63$, $b = 42$, $c = 33$, and $d = 22$.

As the product of extremes = $63 \times 22 = 1386$

And, the product of means = $33 \times 42 = 1386$

So, the product of extremes = the product of means

Hence, 63, 42, 33, and 22 are in proportion.

6.5 Continued Proportion

A continued proportion is one in which the ratio between the first and second term is equal to the ratio between the second and the third term. If a , b and c are in continued proportion, then

$$\frac{a}{b} = \frac{b}{c}$$

Here, a and c are respectively known as the first, and the third terms and b is known as the mean proportional.

We know,

Product of extremes = Product of means

$$b \times b = a \times c$$

$$b^2 = ac$$

$$b = \pm\sqrt{ac}$$

$\therefore$ The mean proportion of a and c is $\sqrt{ac}$.

Examples

(i) Find the mean proportion between 4 and 9 .

Solution: Let the mean proportion be x

Therefore, $4 : x = x : 9$

$$\Rightarrow x \times x = 4 \times 9$$

$$\Rightarrow x^2 = 36$$

$$\Rightarrow x^2 = 6^2$$

$$\Rightarrow x = \pm 6$$

(ii) Find, m , if 7, 14, and m are in continued proportion.

Solution: x, y and z are in continued proportion $xz = y^2$

Let 7, 14, and m be x, y and z respectively.

Therefore, $7m = 14^2$

$$\text{or, } 7m = 196$$

$$\text{or, } m = \frac{196}{7}$$

Therefore, $m = 28$.

Hence, $m = 28$.

6.6 Variation

When two quantities are related such that a change in one causes a change in the other, we say they vary with each other. There are two main types of variation.

Direct Variation (Direct Proportion)

Two quantities x and y are in direct variation if an increase in x causes a proportional increase in y , and a decrease in x causes a proportional decrease in y .

Symbol: $x \propto y$ (x is directly proportional to y)

Real-Life Examples of Direct Variation:

Quantity 1 increases	Quantity 2 increases	Reason
More time spent travelling	More distance covered (at same speed)	Distance $\propto$ Time
More workers hired	More work completed per day	Work $\propto$ Workers
More items purchased	More total cost	Cost $\propto$ Quantity
More principal amount	More interest earned	Interest $\propto$ Principal

Examples

- (i) In which of the following tables, a and b vary directly?
Also, find the constant of proportion if a and b are in direct proportion.

(a)

a	4	7	21	28
b	12	21	63	84

(b)

a	2.5	5	7.5	10	15
b	10	20	30	40	60

(c)

a	1	2	3	4	5
b	2	1	6	3	2/5

Solution: (a) We have, $\frac{4}{12} = \frac{7}{21} = \frac{21}{63} = \frac{28}{84} = \frac{1}{3}$.

Thus, the ratio of the corresponding values of a and b is constant and is equal to $\frac{1}{3}$.

Hence, a and b are in direct proportion with the constant of proportion equal to $\frac{1}{3}$.

(b) We have, $\frac{2.5}{10} = \frac{5}{20} = \frac{7.5}{30} = \frac{10}{40} = \frac{15}{60} = \frac{1}{4}$

This shows that the ratio of the corresponding values of a and b is constant and is equal to $\frac{1}{4}$.

Thus, a and b vary directly. The constant of proportion is $\frac{1}{4}$.

(c) It is evident from the table that the ratio of the corresponding values of a and b is not constant. So, a and b are not in direct proportion.



Practice Questions - 6

- (i) Two numbers are in the ratio 4:9. If the difference between two numbers is 50, what is the larger number?
 (A) 72 (B) 90 (C) 108 (D) 64
- (ii) If Rs. 782 be divided into three parts, proportional to $\frac{1}{2} : \frac{2}{3} : \frac{3}{4}$, then the first part is:
 (A) Rs. 182 (B) Rs. 190 (C) Rs. 196 (D) Rs. 204
- (iii) If 30 dozen eggs cost ₹300. Find the cost of 5 dozen eggs.
 (A) ₹55 (B) ₹45 (C) ₹40 (D) ₹50
- (iv) A sum of money is divided between two persons in the ratio 7:9. If the difference between their shares is ₹240, what is the total sum of money?
 (A) ₹2,880 (B) ₹1,920 (C) ₹3,240 (D) ₹1,980
- (v) Nikki has a bag of 120 coins. The ratio of pennies to nickels to dimes is 2:3:7. How many pennies, nickels and dimes are in her bag?
 (A) 20 pennies, 20 nickels and 80 dimes (B) 10 pennies, 40 nickels and 70 dimes
 (C) 30 pennies, 20 nickels and 70 dimes (D) 20 pennies, 30 nickels and 70 dimes
- (vi) In a mixture, the ratio of alcohol to water is 3:5. If 40 liters of the mixture contains 15 liters of alcohol, how much more alcohol is required to make the ratio of alcohol to water 4:5?
 (A) 5 liters (B) 12 liters (C) 15 liters (D) 20 liter
- (vii) If $0.75 : x :: 5 : 8$, then x is equal to:
 (A) 1.12 (B) 1.2 (C) 1.25 (D) 1.30
- (viii) The continued ratio of Rs.200, Rs.400 and Rs.600 is
 (A) 2 : 4 (B) 4 : 6 (C) 6 : 2:4 (D) 2 : 4 : 6
- (ix) What is the value of x if $\frac{10}{3} : x :: \frac{5}{2} : \frac{5}{4}$?
 (A) 2 (B) 8 (C) $\frac{5}{3}$ (D) $\frac{10}{3}$
- (x) If two cardboard boxes occupy 500 cubic centimetres of space, then how much space is required to keep 200 such boxes?
 (A) 55,000 cubic centimetres (B) 50,000 cubic centimetres
 (C) 45,000 cubic centimetres (D) 35,000 cubic centimetres

7. Fraction & Percentage

7.1 Fractions

A fraction represents a part of a whole or a division of a quantity into equal parts. It consists of two numbers: the numerator and the denominator.

Essential takeaways from types of Fractions:

- (i) The value of proper fraction after further simplification is always less than 1.
- (ii) The simplification of improper fraction results in the value which is equal or greater than 1, but not less than 1.
- (iii) Mixed fractions can always be converted into a fraction.
- (iv) An improper fraction can be converted into a mixed fraction.
- (v) A mixed fraction is always greater than 1.

7.2 Operations on Fractions:

(a) Addition / Subtraction of Fractions:

To add or subtract fractions:

- If necessary, convert the fractions so they have the lowest common denominator.
- Add or subtract the new numerators. The denominator stays the same.

Steps for Adding/subtracting Mixed Fractions:

- (i) Convert the Mixed Fractions to Improper Fractions:
- (ii) Add/subtract the Numerators:
- (iii) simplify it into a mixed fraction or a whole number if necessary.

Examples

(i) Add $3\frac{1}{4} + 2\frac{3}{4}$

Solution: Convert $3\frac{1}{4}$ into an improper fraction:

$$3\frac{1}{4} = \frac{(4 \times 3) + 1}{4} = \frac{12 + 1}{4} = \frac{13}{4}$$

Convert $2\frac{3}{4}$ into an improper fraction:

$$2\frac{3}{4} = \frac{(4 \times 2) + 3}{4} = \frac{8 + 3}{4} = \frac{11}{4}$$

Since both fractions have the same denominator (4), add the numerators:

$$\frac{13}{4} + \frac{11}{4} = \frac{13 + 11}{4} = \frac{24}{4}$$

(b) Multiplication

To multiply two fractions, we multiply the two numerators to get the new numerator, and multiply the two denominators to get the new denominator.

$$\text{Product of two fractions} = \frac{\text{Product of numerators}}{\text{Product of denominators}} = \frac{a}{b} \times \frac{c}{d} = \frac{a \times c}{b \times d}$$

(c) Division of fractions

To divide fractions, flip the second fraction (write its reciprocal) and then multiply. For example, $(3/4) \div (2/5)$ becomes $(3/4) \times (5/2) = 15/8$.

Examples

- (i) **Tenzin drinks $\frac{1}{2}$ glass of milk every day. How many glasses of milk does he drink in a week?**

Solution: Glass of milk drank every day = $\frac{1}{2}$

$$\text{Glasses of milk drank in a week} = \frac{1}{2} \times 7 = \frac{7}{2}$$

Days in month of January = 31

$$\text{Glasses of milk drank in month of January} = 31 \times \frac{1}{2} = \frac{31}{2} = 15\frac{1}{2}.$$

- (ii) **There were 420 students participating in the Swachh Bharat campaign. They cleaned $\frac{42}{75}$ part of the town, Sevagram. What part of Sevagram did each student clean if the work was equally shared by all?**

Solution: 420 students cleaned $\frac{42}{75}$ part of the Sevagram town.

1 student cleaned the part of the Sevagram town equal to $\frac{42}{75} \div 420$

$$\begin{aligned} &= \frac{42}{75} \times \frac{1}{420} \\ &= \frac{42}{75 \times 420} \\ &= \frac{1}{75 \times 10} \\ &= \frac{1}{750} \end{aligned}$$

7.3 Percentage

- Per cent means 'for every hundred'. A fraction in which the denominator is 100, is a percentage. The denominator, i.e., 100 is denoted by a special symbol %, read as the per cent.

For Example: $\frac{10}{100} = 10\%$, $\frac{25}{100} = 25\%$, $\frac{x}{100} = x\%$

(a) Converting Fractional Numbers to Percentages

To convert a fractional number into percentage, multiply the fraction by 100 and put the percentage sign.

Examples

- (i) Express $\frac{7}{20}$ as a percentage.

Solution: $\frac{7}{20} = \frac{7}{20} \times 100\% = 35\%$

(b) Converting Decimals to Percentages

To convert a decimal number into percentage, first, convert it into a fraction, then proceed with the method of converting fractional numbers to percentages.

Examples

- (i) Convert 0.75 into a percentage.

Solution: $0.75 = 0.75 \times 100\%$
 $= \frac{75}{100} \times 100\% = 75\%$

(c) Expressing Percentage as a Fraction or Decimal

Any percentage can be expressed as a decimal or a fraction by dividing the percentage figure by 100.

As $x\% = x$ out of 100 $= \frac{x}{100}$.

So, $75\% = 75$ out of 100 $= \frac{75}{100} = \frac{3}{4}$ or 0.75.

Examples

- (i) Out of 50 students in a class, 15 like to play cricket. What is the percentage of students who like to play cricket?

Solution: Total students = 50

Students who like to play cricket = 15

So, the percentage age of students who like to play cricket $= \frac{15}{50} \times 100\% = 30\%$

- (ii) In a village, 40% of the population comprises of children below 14 years of age. If the ratio of the number of men to the number of women is 2:1, then find the number of women in that village, given that the total population of the village is 5000.

Solution: Total population = 5000

The population of children $= \frac{40}{100} \times 5000 = 2000$

The rest of the population = 3000

Let the number of men and women be 2k and k, respectively.

$\therefore 3k = 3000 \Rightarrow k = 1000$

$\therefore$ Number of women = 1000

(d) Increase or Decrease as Percentage

There are times when we need to know the increase or decrease in a certain quantity as a percentage.

$$\text{Percentage Increase} = \frac{\text{Increased Value} - \text{Original Value}}{\text{Original Value}} \times 100$$

$$\text{Percentage Decrease} = \frac{\text{Original Value} - \text{Decreased Value}}{\text{Original Value}} \times 100$$

Examples

- (i) A school team won 6 games this year against 4 games won last year. What is the percentage increase in the number of wins?

Solution: The increase in the number of wins (or amount of change) = $6 - 4 = 2$.

$$\begin{aligned} \text{Percentage increase} &= \frac{\text{amount of change}}{\text{original amount or base}} \times 100 \\ &= \frac{\text{increase in the number of wins}}{\text{original number of wins}} \times 100 \\ &= \frac{2}{4} \times 100 = 50\% \end{aligned}$$

- (ii) The population of a city decreased from 25,000 to 24,500. Find the percentage decrease.

Solution:

$$\begin{aligned} \text{Percentage decrease} &= \frac{\text{Decrease in population}}{\text{Initial population}} \times 100 \\ &= \frac{25000 - 24500}{25000} \\ &= \left(\frac{500}{25000} \times 100 \right) \% = 2\% \end{aligned}$$



Practice Questions - 7

- (i) What fraction of an hour is 20 minutes?
 (A) $\frac{1}{10}$ (B) $\frac{1}{3}$ (C) $\frac{2}{5}$ (D) $\frac{7}{13}$
- (ii) Write the following as per cent.
 (a) $\frac{1}{4}$ (b) $\frac{22}{44}$ (c) $\frac{4}{25}$
- (iii) Convert the following percentages into fractions.
 (a) 45% (b) 65% (c) 42.5%
- (iv) William travelled a distance of 10 km. He covered 70% of the distance by bus and the remaining on foot. What distance did he travel by bus? How much distance did he cover on foot?
- (v) Suresh bought $2\frac{1}{2}$ kg sugar whereas mohit bought $3\frac{1}{2}$ kg of sugar. Find the total amount of sugar bought by both of them.
- (vi) Viraj is $6\frac{1}{6}$ feet tall. 4 years ago, he was $1\frac{1}{4}$ feet shorter. How tall was Viraj four years ago?
- (vii) Safia saw the Moon setting on Monday at 10 pm. Her mother, who is a scientist, told her that every day the Moon sets $\frac{5}{6}$ hour later than the previous day. How many hours after 10 pm will the moon set on Thursday?
- (viii) Manish spends 17% of his monthly income in travelling. He spends 25% of his monthly income on household expenses and spends 36% of his monthly income on families medical expenses. He has remaining amount of Rs. 10032 as cash with him. What is Manish's annual income?
- (ix) The population of a city in the year 2018 was 72,400 and in 2019, it was 76,925 . Find the percentage increase in the population.
- (x) A shopkeeper expects a gain of 22.5% on his cost price. If in a week, his sale was of Rs. 392, what was his profit?

8. PROFIT AND LOSS

When a person buys an article for a certain price and then sells it for a different price, he makes a profit or incurs a loss.

8.1 Basic Terms

Cost Price (C.P.): The price at which an article is purchased is called its cost price.

Selling Price (S.P.): The price at which an article is sold is called its selling price.

Marked Price (M.P.): In big shops and departmental stores, every article is tagged with a card and its price is written on it. This is called the marked price of that article, abbreviated as M.P.

Discount: It is the reduction in the price of goods or services offered by shopkeepers at the marked price to increase the sale or clear the old stock.

Note: The discount is always calculated on the marked price.

Selling price = (marked price) – (discount)

8.2 Profit and Loss

In case of profit	In case of Loss
Profit = S.P. – C.P.	Loss = C.P. – S.P.
S.P. = Profit + C.P.	C.P. = Loss + S.P.
C.P. = S.P. – Profit	S.P. = C.P. – Loss

Examples

- (i) A milkman buys 20 litres of milk from a dairy for ₹370. He sells it at the rate of ₹21.50 per litre. Find his profit or loss.

Solution: C.P. of 20-litre milk = ₹370

S.P. of 1-litre milk = ₹21.50

Therefore, S.P. of 20 litres of milk = ₹21.50 × 20 = ₹430

S.P. > C.P.,

So profit = ₹430 – ₹370 = ₹60

8.3 Profit or Loss Percentage

To calculate the profit or loss in percentage, we use the following formulae:

$$\text{Profit \%} = \frac{\text{Profit}}{\text{C.P.}} \times 100$$

$$\text{Loss \%} = \frac{\text{Loss}}{\text{C.P.}} \times 100$$

Profit or loss per cent is always calculated on the C.P.

$$(a) \text{ S.P.} = \frac{\text{C.P.} \times (100 + \text{Profit \%})}{100}$$

In case of profit %

$$\text{S.P.} = \frac{\text{C.P.} \times (100 - \text{Loss \%})}{100}$$

In case of loss %

$$(b) \text{ C.P.} = \frac{\text{S.P.} \times 100}{(100 + \text{Profit}\%)}$$

In case of profit %

$$\text{C.P.} = \frac{\text{S.P.} \times 100}{(100 - \text{Loss}\%)}$$

In case of loss %

Examples

- (i) **Krishna sold a motorbike for ₹15000, losing 25%. Find the cost price of a motorbike.**

Solution: S.P. = (100 - loss%)% of C.P.

$$\Rightarrow 15,000 = \left[\frac{(100 - 25)}{100} \right] \times \text{C.P.}$$

$$\Rightarrow \text{C.P.} = 15,000 \times \frac{100}{75} \Rightarrow \text{C.P.} = ₹20,000$$

- (ii) **Nandan sells a quintal of wheat for ₹308, thereby, gaining a profit of 12%. By selling a quintal of rice for the same amount, he losses 12%. Find the C.P. of both rice and wheat. In addition, find his total gain or loss.**

Solution: Given S.P. of wheat = ₹308, Gain = 12%

we know, $\text{C.P.} = \frac{\text{S.P.} \times 100}{(100 + \text{gain}\%)}$

Therefore, $\text{C.P.} = \frac{308 \times 100}{100 + 12} = \frac{308 \times 100}{112} = ₹275$

Now, S.P. of rice = ₹308

Loss = 12%

We know, $\text{C.P.} = \frac{100 \times \text{S.P.}}{(100 - \text{Loss}\%)}$

Therefore, $\text{C.P. of rice} = \frac{100 \times 308}{(100 - 12)} = \frac{100 \times 308}{88} = ₹350$

Total C.P. of wheat and rice = ₹(275 + 350) = ₹625

Total S.P. = ₹(308 × 2) = ₹616

We can see that S.P. < C.P.

$\therefore \text{Loss} = ₹625 - ₹616 = ₹9$

- (iii) **A man purchases two fans for Rs. 2160. By selling one fan at a profit of 15% and the other at a loss of 9%, he neither gains nor losses in the whole transaction. Find the cost price of each fan.**

Solution: Let, Cost price of the first fan = Rs. x

Cost price of the second fan = Rs. 2160 - x

According to the question,

The gain on selling the first fan is 15% of its cost price = 0.15x

The loss on selling the second fan is 9% of its cost price = 0.09 × (2160 - x)

Therefore,

$$\Rightarrow 0.15x = 0.09(2160 - x)$$

$$\Rightarrow 0.15x = 194.4 - 0.09x$$

$$\Rightarrow 0.24x = 194.4 \Rightarrow x = 810$$

Cost price of the second fan = 2160 - 810 = Rs. 1350

$\therefore$ The cost price of the first fan is Rs. 810 and the cost price of the second fan is Rs. 1350.

- (iv) **A man sells two articles at Rs. 9,180 each. He gains 8% on one article and loses 15% on the other. Find his overall profit or loss.**

Solution: Let the cost price of the first and second article be P and Q.

According to the question,

$$\Rightarrow P + P \times 8\% = 9180 \Rightarrow P + 0.08P = 9180 \Rightarrow 1.08P = 9180$$

$$\Rightarrow P = \frac{9180}{1.08} \Rightarrow P = 8500$$

$$\text{Now, } Q - Q \times 15\% = 9180 \Rightarrow Q - 0.15Q = 9180$$

$$\Rightarrow 0.85Q = 9180 \Rightarrow Q = \frac{9180}{0.85}$$

$$\Rightarrow Q = 10,800$$

$$\text{Total CP} = 10,800 + 8500 = \text{Rs. } 19,300$$

$$\text{Total SP} = 9180 \times 2 = \text{Rs. } 18,360$$

Now, the overall loss;

$$\Rightarrow 19,300 - 18,360 = \text{Rs. } 940$$

$\therefore$ His overall loss is Rs. 940.

8.4 Simple Interest

Interest is the money paid by a borrower to the lender for using the money for a specified time period.

$$\text{Simple Interest (S.I.)} = \frac{P \times T \times R}{100}$$

$$\text{Amount} = P + \text{SI}$$

Examples

- (i) **Find the simple interest on ₹ 2560 for 3 years at 15% per annum.**

Solution: Principal (P) = ₹2560, Time period (T) = 3 years, Rate(R) = 15% p.a.

$$\text{Simple interest (I)} = \frac{PTR}{100}$$

$$= 2560 \times 3 \times \frac{15}{100} = ₹1152$$

- (ii) **At what rate of percent, ₹1500 amounts to ₹2100 in 4 years?**

Solution: Time period (T) = 4 years, Amount (A) = ₹2100, Principal (P) = ₹1500

$$\text{Interest} = A - P$$

$$= ₹(2100 - 1500) = ₹600$$

$$\text{We have, S.I.} = \frac{PTR}{100}$$

$$600 = 1500 \times 4 \times \frac{R}{100}$$

$$\Rightarrow 600 = 60 \times R$$

$$\Rightarrow R = \frac{600}{60} \Rightarrow R = 10$$

$\therefore$ Rate of interest = 10% p.a.

- (iii) **Given that simple interest on a certain sum of money is Rs. 4016.25 at 9% per annum in 5 years. Find the sum of money.**

Solution: Let P be the principal amount or sum of money.

Time (T) = 5 years

Rate of interest (R) = 9%

Simple interest earned (SI) = Rs. 4016.25

As we know,

$$SI = PTR/100$$

$$\text{Rs. } 4016.25 = (P \times 5 \times 9)/100$$

$$P = (\text{Rs. } 4016.25 \times 100)/(5 \times 9)$$

$$= \text{Rs. } 8925$$

Therefore, the sum of money is Rs. 8925.

- (iv) **A sum of Rs. 25000 will become Rs. 31000 in 48 months at some rate of simple interest. Find the rate of interest per annum.**

Solution: Given,

Principal amount (P) = Rs. 25000

Time (T) = 48 months = $(48/12)$ years = 4 years

Total amount after 4 years (A) = Rs. 31000

Let R be the rate of interest.

As we know,

$$A = P + SI$$

$$\text{Rs. } 31000 = \text{Rs. } 25000 + [(\text{Rs. } 25000 \times 4 \times R)/100]$$

$$\text{Rs. } 31000 - \text{Rs. } 25000 = \text{Rs. } 1000 \times R$$

$$R = \text{Rs. } 6000 / \text{Rs. } 1000 = 6$$

Therefore, the rate of interest per annum is 6%.



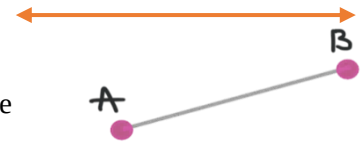
Practice Questions - 8

- (i) A trader purchased 10 quintals of wheat from a farmer for Rs. 8,750. He sold it at Rs. 11.50 per kg. Find the amount of profit/ loss made by the trader.
- (ii) Karim bought 150 dozen pencils at Rs. 20 a dozen. He sold them at Rs. 2.50 per pencil. Find the profit or loss per cent.
- (iii) Neelima bought 2400 bananas at Rs. 15 a dozen. She sold 1350 of them at Rs. 4 for 2 and the remaining at Rs. 8 for 5. Find her gain or loss percentage.
- (iv) At what rate of interest, ₹2500 amounts to ₹2800 in 2 years?
- (v) A principal amount becomes $\frac{5}{3}$ of itself in 8 years at a certain rate of interest. Find the rate of interest.
- (vi) A merchant sells two items each for ₹1980. On one he gains 10% and on the other he loses 10%. His overall gain or loss% is:
- (A) No profit, no loss (B) 1% profit
(C) 1% loss (D) 2% loss
- (vii) If the C.P. of 15 articles equals the S.P. of 12 articles, the profit% is:
- (A) 20% (B) 25%
(C) 30% (D) 33.33%
- (viii) A shopkeeper marks goods 40% above C.P. and gives 25% discount. His profit% is:
- (A) 5% (B) 10%
(C) 12% (D) 15%
- (ix) After selling an article at a loss of 15%, Ravi found that had he sold it for ₹99 more, he would have gained 8%. The C.P. of the article is:
- (A) ₹165 (B) ₹170
(C) ₹175 (D) ₹180
- (x) A dealer allows a discount of 10% and still makes a profit of 8%. If C.P. is ₹100, find the Marked Price:
- (A) ₹110 (B) ₹115
(C) ₹120 (D) ₹125

9. LINES AND ANGLES

9.1 Line

(a) Line: A line is a set of points or a combination of points in a straight path that extends in opposite directions without an end. It has no definite length.



(b) Line Segment: A line segment is part of a line having two endpoints. Here, in the figure, AB is a line segment. It has a definite length.

(c) Ray: A ray is part of a line having one endpoint i.e. one end of the line is fixed and it can be extended indefinitely in the opposite direction. Here, P is the endpoint (fixed).



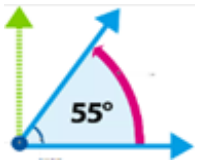
9.2 Angles

Angles are geometric figures formed by two rays, meeting at a common endpoint. The two rays are called the sides of the angle and the common endpoint is called the vertex. Angles are generally measured in degrees.

9.3 Types of Angles

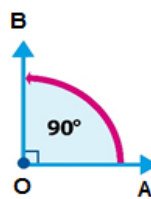
(a) Acute Angle

An angle whose measure is between 0° and 90° is called an acute angle. In the figure, $\angle AOB$ is an acute angle.



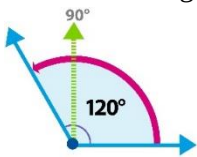
(b) Right Angle

An angle of measure 90° is called a right angle. In the figure, $\angle BOA$ is a right angle.



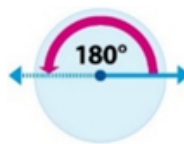
(c) Obtuse Angle

An angle whose measure is more than 90° but less than 180° is called an obtuse angle as shown in the figure.



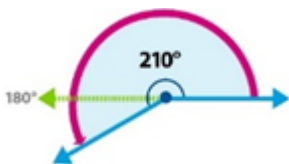
(d) Straight Angle

An angle of 180° is called a straight angle as shown in the figure.



(e) Reflex Angle:

An angle whose measure is greater than 180° and less than 360°



(f) Complete Angle

An angle of 360° is called a complete angle as shown in the figure.



9.4 Pair of Angles

(a) Complementary Angles: Two angles are said to be complementary if the sum of their measures is 90° (if $\angle A + \angle B = 90^\circ$) then angle A and angle B are called the complement of each other.

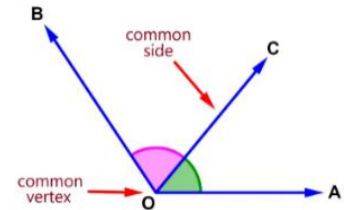
For example: $(45^\circ, 45^\circ), (10^\circ, 80^\circ), (20^\circ, 70^\circ), (30^\circ, 60^\circ)$

(b) Supplementary Angles: Two angles are said to be supplementary if the sum of their measures is 180° i.e. (if $\angle A + \angle B = 180^\circ$) and angle A and angle B are called the supplement of each other.

For example: $(10^\circ, 170^\circ), (20^\circ, 160^\circ), (30^\circ, 150^\circ), (40^\circ, 140^\circ), (50^\circ, 130^\circ)$ etc. are all pairs of supplementary angles.

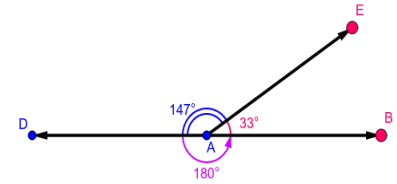
(c) Adjacent Angles: Two angles are called adjacent angles if:

- (i) they have a common vertex (in figure point O is a common vertex).
 - (ii) they have a common arm (in the figure, OC is a common arm).
 - (iii) their other arms lie on either side of the common arm (in the figure, OA and OB)
- So, in the figure $\angle BOC$ and $\angle AOC$ are adjacent angles.



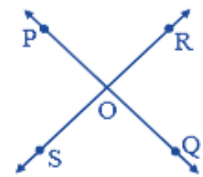
(d) Linear Pair Angles: Two adjacent angles are said to form a linear pair if their non-common arms are two opposite rays.

If figure $\angle DAE$ and $\angle BAE$ form a linear pair as their non-common arms AD and AB are two opposite rays i.e., DAB is a line.



(e) Vertically Opposite Angles: When two lines intersect, the angles that are formed opposite to each other at the point of intersection (vertex) are called vertically opposite angles.

Note: Vertically opposite angles are always equal.



Important Facts:

- The sum of all angles formed on the same side of a line at a given point on a line is 180° .
- The sum of all angles around a point is 360° .

Examples

(i) Find the complement of the given angles.

Solution: Complement of $20^\circ = 90^\circ - 20^\circ = 70^\circ$. ($\because \angle A + \angle B = 90^\circ$)

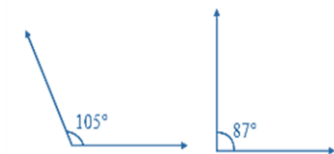
Complement of $63^\circ = 90^\circ - 63^\circ = 27^\circ$.



(ii) Find the supplement of each of the given angles.

Solution: Supplement of $105^\circ = 180^\circ - 105^\circ = 75^\circ$.

Supplement of $87^\circ = 180^\circ - 87^\circ = 93^\circ$.



(iii) Find the angle which is equal to its complement.

Solution: Let the angle = x°

The complement of $x^\circ = 90^\circ - x^\circ$

According to the question, $x^\circ = 90^\circ - x^\circ$

$$\Rightarrow x^\circ + x^\circ = 90^\circ \Rightarrow x^\circ = 45^\circ$$

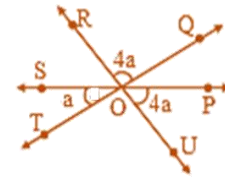
(iv) Determine the value of a.

Solution: Since $\angle ROQ$ and $\angle TOU$ are vertically opposite angles.

So $\angle TOU = \angle ROQ \Rightarrow \angle TOU = 4a$ ($\angle ROQ = 4a$ given)

$\therefore \angle SOT + \angle TOU + \angle UOP = 180^\circ$ (angles on a straight line)

$$a + 4a + 4a = 180^\circ \Rightarrow 9a = 180^\circ \Rightarrow a = 20^\circ$$

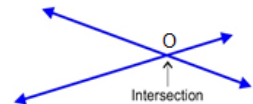


9.5 Pair of Lines

Pair of lines implies the lines that are related or connected to each other like intersecting lines, perpendicular lines and parallel lines.

(a) **Intersecting Lines**

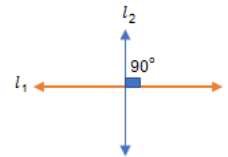
Two lines are said to be intersecting if they have a point in common. This common point is called the point of intersection of intersecting lines.



(b) **Perpendicular Lines**

Two lines are said to be perpendicular to each other if they intersect at 90° angle (right angle).

In figure l_1 and l_2 are perpendicular lines i.e., $l_1 \perp l_2$.



(c) **Parallel Lines**

Two coplanar lines l and m are said to be parallel if:

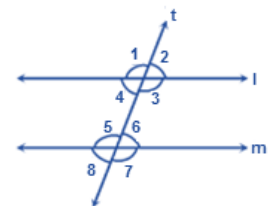
- (i) They lie in the same plane
- (ii) The perpendicular distance between them is always equal.



9.6 Transversal of Parallel Lines

When transversal intersects two parallel lines, then

- Pair of alternate interior angles are equal i.e., $\angle 4 = \angle 6, \angle 3 = \angle 5$
- Pair of alternate exterior angles are equal i.e., $\angle 1 = \angle 7, \angle 2 = \angle 8$
- Pair of corresponding angles are equal i.e., $\angle 1 = \angle 5, \angle 2 = \angle 6, \angle 4 = \angle 8, \angle 3 = \angle 7$
- Pair of co-interior angles are supplementary i.e., $\angle 4 + \angle 5 = 180^\circ, \angle 3 + \angle 6 = 180^\circ$



Examples

- (i) In Fig., AB and CD are parallel lines intersected by a transversal PQ at L and M respectively. If $\angle LMD = 35^\circ$ find $\angle ALM$ and $\angle PLA$.

Solution: Given that, $\angle LMD = 35^\circ$

From the figure, we can write,

$\angle LMD$ and $\angle LMC$ are forming a linear pair

$$\angle LMD + \angle LMC = 180^\circ \quad [\text{sum of angles in linear pair} = 180^\circ]$$

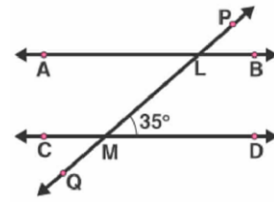
On rearranging, we get

$$\angle LMC = 180^\circ - 35^\circ = 145^\circ$$

So, $\angle LMC = \angle PLA = 145^\circ$ [Corresponding angles]

$\angle ALM = \angle LMD = 35^\circ$ [alternate interior angles]

Therefore, $\angle ALM = 35^\circ$, $\angle PLA = 145^\circ$.



- (ii) In Fig., $l \parallel m$ and $p \parallel q$. Find the values of x, y, z, t .

Solution: Given that one of the angles is 80° .

$\angle z$ and 80° are vertically opposite angles

Therefore $\angle z = 80^\circ$

$\angle z$ and $\angle t$ are corresponding angles

$$\angle z = \angle t$$

Therefore, $\angle t = 80^\circ$

$\angle z$ and $\angle y$ are corresponding angles

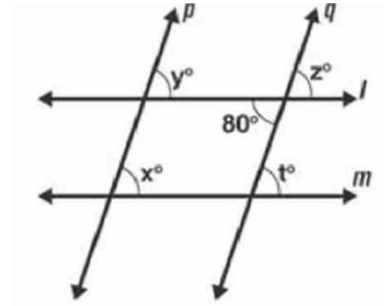
$$\angle z = \angle y$$

Therefore, $\angle y = 80^\circ$

$\angle x$ and $\angle y$ are corresponding angles

$$\angle y = \angle x$$

Therefore, $\angle x = 80^\circ$



- (iii) In the given figure, if $AB \parallel CD$, $\angle APQ = 50^\circ$ and $\angle PRD = 130^\circ$, then find $\angle QPR$.

Solution: Since AB and CD are parallel and PR is a transversal.

$$\therefore \angle BPR + \angle PRD = 180^\circ$$

[$\because$ the sum of consecutive interior angles is 180°]

$$\Rightarrow \angle BPR + 130^\circ = 180^\circ$$

$$\Rightarrow \angle BPR = 180^\circ - 130^\circ$$

$$\Rightarrow \angle BPR = 50^\circ$$

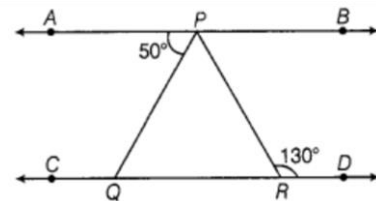
$$\text{Also, } \angle APQ + \angle QPR + \angle BPR = 180^\circ$$

[$\because$ the sum of all the angles on a straight line is 180°]

$$\Rightarrow 50^\circ + \angle QPR + 50^\circ = 180^\circ$$

$$\Rightarrow \angle QPR + 100^\circ = 180^\circ$$

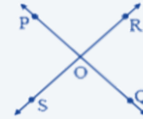
$$\Rightarrow \angle QPR = 180^\circ - 100^\circ \therefore \angle QPR = 80^\circ$$



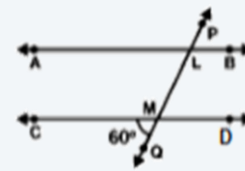
Practice Questions - 9



- (i) Two angles forming a linear pair are _____.
- (ii) The difference in the measures of two complementary angles is 12° . Find the measures of the angles.
- (iii) Find the angle, which is equal to its supplement.
- (iv) Two lines PQ and RS intersect at O. If $\angle POR = 50^\circ$, then find the value of $\angle ROQ$.

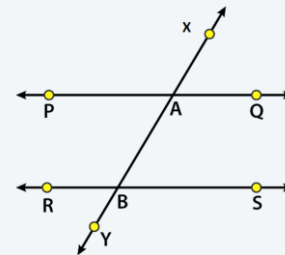


- (v) In Figure, AB and CD are parallel lines intersected by a transversal PQ at L and M respectively, If $\angle CMQ = 60^\circ$, find $\angle PLB$.

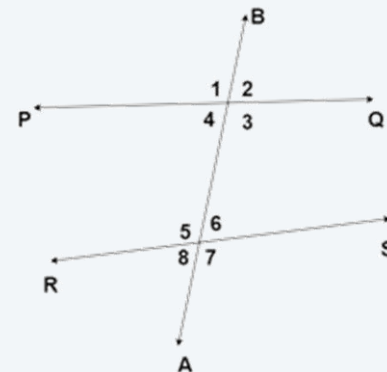


- (vi) A transversal cuts two parallel lines at two different points. If x and y are pair of interior angles on the same side of the transversal and x is equal to y, then what are the angles?
- (vii) A transversal cuts two parallel lines at two different points. If one of the angles is 45° , then its corresponding angle will be?
- (viii) A transversal cuts two parallel lines at two different points. If one of the angles is 108° , then its vertically opposite angle is?
- (ix) Match column A and column B with the help of Fig.

Column A	Column B
(i) Vertically opposite angles	(i) $\angle PAB$ and $\angle ABS$
(ii) Alternate angles	(ii) $\angle PAB$ and $\angle RBY$
(iii) Corresponding angles	(iii) $\angle PAB$ and $\angle XAQ$



- (x) From the given diagram, if a transversal intersects two parallel lines, then what is the relation between pair of alternate interior angles $\angle 4$ and $\angle 5$?
 - (A) $\angle 4 + \angle 5 = 180^\circ$
 - (B) $\angle 4 + \angle 5 = 90^\circ$
 - (C) $\angle 4 = \angle 5$
 - (D) $\angle 4 > \angle 5$



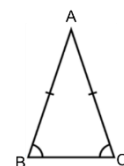
10. THE TRIANGLE & ITS PROPERT

10.1 Types of Triangles

(a) Based on their sides

(i) Equilateral Triangle: An Equilateral triangle is a triangle in which all three sides are equal and angles are also equal. The value of each angle of an equilateral triangle is 60° .

For Example: In $\triangle ABC$, $AB = BC = CA$, $\angle A = \angle B = \angle C = 60^\circ$

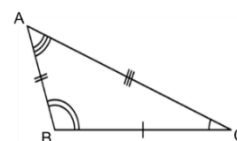


(ii) Isosceles Triangle: An Isosceles triangle is a triangle that has two equal sides. Also, the two angles opposite to the two equal sides are equal.

For Example: In $\triangle ABC$, $AB = BC$, $\angle ABC = \angle ACB$

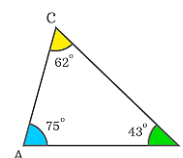
(iii) Scalene triangle: A scalene triangle is a triangle in which all three sides are in different lengths, and all three angles are of different measures.

For Example: In $\triangle ABC$, $AB \neq BC \neq CA$, $\angle A \neq \angle B \neq \angle C$

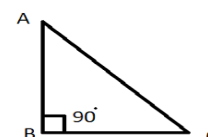


(b) Based on their angles

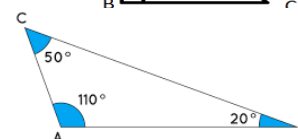
(i) Acute-Angled Triangle: A triangle in which all the angles are acute (less than 90°) is an acute-angled triangle. In the given triangle, each of the angles A, B, and C is less than 90° . Hence, $\triangle ABC$ is an acute-angled triangle.



(ii) Right-Angled Triangle: A triangle which has a right angle is a right-angled triangle. In the right triangle, $\angle ABC = 90^\circ$.



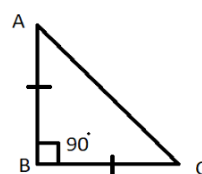
(iii) Obtuse-Angled Triangle: A triangle in which one angle is greater than 90° is an obtuse-angled triangle.



(iv) Isosceles Right-Angled Triangle: A triangle in which two sides are equal and one angle is 90° is an isosceles right-angled triangle.

In the triangle, ABC , $AB = BC$ and $\angle ABC = 90^\circ$.

Hence, $\triangle ABC$ is an Isosceles Right-Angled Triangle.



10.2 Angle Sum Property of Triangle

Theorem: The sum of the three angles of a triangle is 180° .

Given: ABC is a triangle.

To prove: $\angle A + \angle B + \angle C = 180^\circ$

Construction: Draw a line XY through A and parallel to BC .

Proof: $\angle YAB + \angle BAC + \angle XAC = 180^\circ$ ($\because$ straight line) ... (1)

XY and BC are parallel and AC is a transversal.

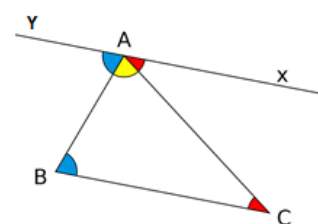
$\therefore \angle XAC = \angle ACB$ ($\because$ alternate angles) ... (2)

XY and BC are parallel and AB is a transversal.

$\therefore \angle YAB = \angle ABC$ ($\because$ alternate angles) ... (3)

From Eqn (1), (2), and (3), we have $\angle ABC + \angle BAC + \angle ACB = 180^\circ$

Hence proved.



10.3 Exterior Angle Property of a Triangle

If we extend any side of a triangle, the angle that is formed with this side and its adjacent side is called the exterior angle of a triangle. There are three exterior angles in a triangle.

In the given triangle ABC, AC is produced and then the angle formed ($\angle BCD$) is the exterior angle.

Theorem: The exterior angle of a triangle is equal to the sum of the interior angles opposite to it.

Given: ABC is a triangle; AC is produced to point D.

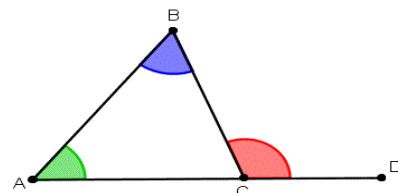
To Prove: $\angle BCD = \angle A + \angle B$

Proof: $\angle A + \angle B + \angle BCA = 180^\circ$ ($\because$ Angles of $\triangle ABC$)

$\angle BCA + \angle BCD = 180^\circ$ ($\because$ Linear pair)

$\therefore \angle A + \angle B + \angle BCA = \angle BCA + \angle BCD \Rightarrow \angle BCD = \angle A + \angle B$

Hence proved.

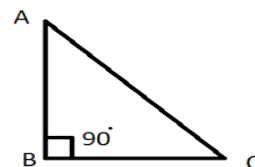


10.4 Pythagoras Theorem

Theorem: In a right-angled triangle $\triangle ABC$, the square of the hypotenuse AC is equal to the sum of the squares of the perpendicular BC and the base AB.

i.e.,

$$AC^2 = AB^2 + BC^2$$



Examples

(i) The sides of a $\triangle ABC$ measure 7 cm, 24 cm and 25 cm. What type of triangle is ABC?

Solution: Since no two sides are equal, ABC is a scalene triangle.

Further, $7^2 + 24^2 = 25^2$,

Since the square of the longest side is equal to the sum of the squares of the other two sides, ABC is a right-angled triangle.

(ii) In $\triangle PQR$, $\angle P = 50^\circ$ and $\angle Q = 60^\circ$. Find $\angle R$

Solution: In a triangle, the sum of the angles is equal to 180° . $\angle P + \angle Q + \angle R = 180^\circ$

$50^\circ + 60^\circ + \angle R = 180^\circ \Rightarrow \angle R = 70^\circ$

(iii) In $\triangle ABC$, $AC = BC$ and $\angle BAC = 50^\circ$. Find $\angle BCA$.

Solution: Given: $AC = BC$

In a triangle, angles opposite to equal sides are equal.

$\therefore \angle ABC = \angle CAB = 50^\circ$

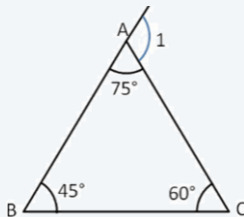
$\angle ABC + \angle BCA + \angle CAB = 180^\circ$

$50^\circ + \angle BCA + 50^\circ = 180^\circ \Rightarrow \angle BCA = 80^\circ$

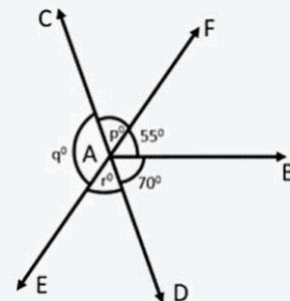


Practice Questions - 10

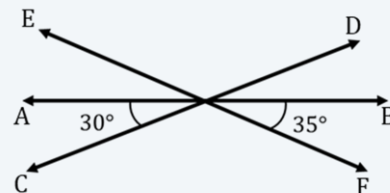
- (i) Which of the following statements is incorrect?
- (A) The sum of any two sides of a triangle is greater than the third side.
 (B) A triangle can have all its angle acute.
 (C) A right-angled triangle cannot be equilateral.
 (D) Difference of any two sides of a triangle is greater than the third side.
- (ii) What type of triangle is $\triangle ABC$, where $\angle B = \angle C = 71^\circ$?
- (A) Equilateral (B) Isosceles (C) Scalene (D) Right-angled
- (iii) Two angles of a triangle measure 90° and 30° . The measure of the third angle is
- (A) 90° (B) 60° (C) 20° (D) 45°
- (iv) The ratio of the measures of the three angles of a triangle is $2 : 3 : 4$. The measure of the largest angle is
- (A) 80° (B) 60° (C) 40° (D) 180°
- (v) Find exterior angle $\angle 1$ in the given figure.



- (vi) The sides of a right triangle are 20 cm and 21 cm, find its hypotenuse.
- (vii) In the given diagram if AB, CD, and EF are straight lines, then $2p + r =$
- (A) 120° (B) 160°
 (C) 165° (D) 240°

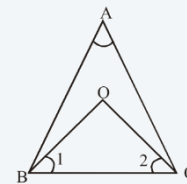


- (viii) In the figure, lines AB, CD and EF intersect at O. Find the measures of $\angle AOE + \angle EOD$.
- (A) 30° (B) 45°
 (C) 50° (D) None



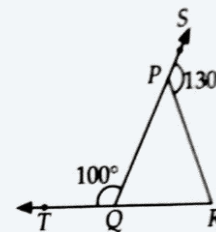
- (ix) In the given figure, BO and CO are respectively the bisectors of $\angle ABC$ and $\angle ACB$, then $\angle BOC$ is equal to

- (A) $90 - \frac{1}{2}\angle A$ (B) $180 - \angle 1 + \angle 2$
 (C) $90 + \frac{1}{2}\angle A$ (D) $180 - (\angle 1 + \angle 2)$



- (x) In the given figure, sides QP and RQ of $\triangle PQR$ are produced to point S and T respectively. If $\angle PQT = 100^\circ$ and $\angle SPR = 130^\circ$, find $\angle PRQ$.

- (A) 45° (B) 35°
 (C) 50° (D) 135°



11. PERIMETER AND AREA

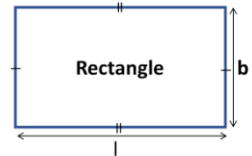
11.1 Area and Perimeter of Plane Figures

For finding the measure of area and perimeter of plane figures we should know about various units used here.

Units of length or distance and the units of area of plane figures are as follows

Squares and Rectangles

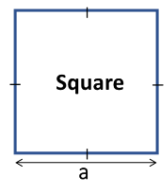
Let us suppose the length of the rectangle is 'l' and its breadth is 'b' then,



- The perimeter of the rectangle = $2 \times (l + b)$ units
- The area of the rectangle = $(l \times b)$ square units

Now, let us take a square of side 'a' then,

- The perimeter of the square = $4a$ units
- The area of the square = $(a \times a) = a^2$ square units



Examples

- (i) Each side of a square is 2.6 cm. Find its perimeter and area.

Solution: Given that,

Side of the square = 2.6 cm

Perimeter of the square = $4 \times \text{side length}$

$$P = 4 \times 2.6 = 10.4 \text{ cm}$$

Area of the square = side^2

$$= (2.6)^2 = 2.6 \times 2.6$$

$$= 6.76 \text{ cm}^2$$

$\therefore$ The perimeter of the square is 10.4 cm and the area is 6.76 cm^2 .

- (ii) Find the area of a square park whose perimeter is 320 m.

Solution: Given: Perimeter of square park = 320 m

$$\Rightarrow 4 \times \text{side} = 320 \text{ m}$$

$$\Rightarrow \text{Side} = \frac{320}{4} = 80 \text{ m}$$

Now, the area of square park = $\text{side} \times \text{side} = 80 \times 80 = 6400 \text{ m}^2$

Thus, the area of square park is 6400 m^2 .

- (iii) Find the perimeter and area of the rectangle of length 17 cm and breadth 13 cm.

Solution: Given,

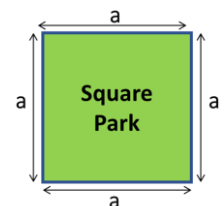
Length of rectangle = 17 cm

breadth of rectangle = 13 cm

Perimeter of rectangle = $2 (\text{length} + \text{breadth})$

$$= 2(17 + 13) = 2(30) = 60 \text{ cm}$$

Area of rectangle = $\text{Length} \times \text{Breadth} = 17 \times 13 = 221 \text{ sq. cm}$



- (iv) A room is 13 m long and 9 m broad. Find the cost of carpeting the room at the rate of Rupees 10 per square meter.

Solution: Given:

Length of the room = 13 m

Breadth of the room = 9 m

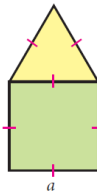
$$\begin{aligned}\text{Area of the room} &= (\text{length} \times \text{breadth}) \\ &= (13 \times 9)\text{m}^2 = 117 \text{ m}^2\end{aligned}$$

$$\text{Cost of carpeting} = 117 \times 10 = \text{Rs. } 1170$$

11.2 Combination of Shapes



Combination of
Rectangle and Semicircle



Combination of square
and Triangle

The area of a shape is the amount of space that is covered by the shape. To find the area of a combination of shapes, the combined shape must be broken up into simple shapes. In this way, the area of each shape can be found using formulas. The individual areas then can be added together to calculate the total area of the combined shape.

Examples

- (i) Find the area of the composite shape which is formed by joining a square and a triangle. The length of the side of the square is 5 units. The base and height of the triangle are 6 units and 7 units respectively.

Solution: Given,

Length of the side of the square = 5 units

Base of the triangle = 6 units, and the height of the triangle = 7 units

Area of composite shape = Area of square + Area of triangle

$$\Rightarrow A = (5)^2 + \left[\left(\frac{1}{2} \right) \times 6 \times 7 \right]$$

$$\Rightarrow A = 25 + 21 = 46 \text{ square units}$$

The area of the composite shape is 46 square units.

- (ii) Calculate the area of this composite shape which is given in the figure below.

Solution: The length and breadth of rectangle ABCD are 2 in and 7 in.

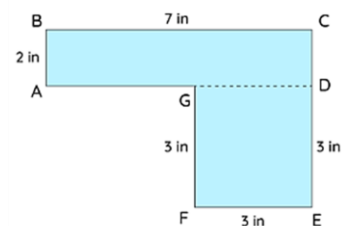
The length of the side of the square DEFG is 3 in.

Using the formula for the area of the composite shape,

Area of composite shape = Area of rectangle + area of the square.

$$\begin{aligned}\Rightarrow \text{Area of composite shape} &= \text{Length} \times \text{Breadth} + \text{side}^2 \\ &= BC \times AB + DE^2 \\ &= 7 \times 2 + 3^2 \\ &= 14 + 9 = 23 \text{ square inches.}\end{aligned}$$

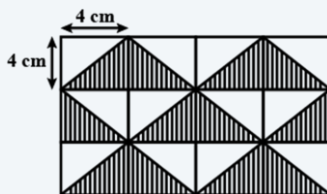
Therefore, the area of the given composite shape is 23 square inches.





Practice Questions - 11

- (i) The area of a square plot is 1600 m^2 . The side of the plot is:
 (A) 40 m (B) 80 m (C) 120 m (D) 160 m
- (ii) The area of a rectangular field is 250 m^2 . If the breadth of the field is 10 m, find its length.
 (A) 25 m (B) 50 m (C) 100 m (D) 125 m
- (iii) The length and breadth of a rectangular field are in the ratio 5 : 3 and its perimeter is 480 m. The area of the field is
 (A) 7200 m^2 (B) 13500 m^2 (C) 15000 m^2 (D) 54000 m^2
- (iv) The difference between the length and breadth of a rectangle is 23 m. If the perimeter is 206 m, then the area is
 (A) 1520 m^2 (B) 2520 m^2 (C) 2420 m^2 (D) None of these
- (v) The perimeter of a square sheet is same as a rectangular sheet. If the side of the square sheet is 30 cm and the length of the rectangular sheet is 45 cm, then area of the rectangular sheet is
 (A) 600 cm^2 (B) 675 cm^2 (C) 240 cm^2 (D) 800 cm^2
- (vi) The area of an isosceles triangle with equal sides 5 meters and the third side 8 meters is
 (A) 12 m^2 (B) 2.86 m^2 (C) 10.45 m^2 (D) None of these
- (vii) A lawn is in the shape of a rectangle of length 80 m and width 40 m. Outside the lawn there is a footpath of uniform width 3 m. Find the area of the path.
 (A) 764 m^2 (B) 789 m^2 (C) 756 m^2 (D) 742 m^2
- (viii) A door of length 1 m and breadth 0.5 m is on a wall. The length of the wall is 4.5 m and the breadth is 3.6 m as shown in Fig. Find the cost of white washing the wall, if the rate of white washing the wall is Rs 20 per m^2 ?
- (ix) The given rectangle is made up of 12 squares. Find the area of the shaded region.



- (x) A lawn is in the shape of rectangle of length 60 m and width 40 m. Inside the lawn there is a footpath of uniform width 1 m bordering the lawn. What is the area of the path?

12. VISUALISING SOLID SHAPES

Basic Concepts

12.1 Plane Figures (2-D shapes)

The figures which can be fully defined by only two dimensions, i.e., length and breadth are called 2D-shapes or plane figures. Triangles, rectangles, squares, circles etc. can be drawn on paper, so these are plane figures. The plane figures extend in two directions which are perpendicular to each other.

12.2 Solids (3-D shapes)

The shapes and figures which require three dimensions i.e., length, breadth and height or depth to be completely defined are called solid figures or 3-dimensional figures.

Objects which occupy space are called solids. A solid occupies a fixed amount of space and has three dimensions.

(a) Cuboids

A solid bounded by six rectangular faces (not all equal) is called a cuboid. It has three dimensions, namely length, breadth and height.

Various parts of a cuboid are

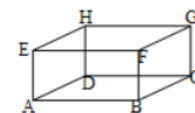
(i) Face: A cuboid has 6 rectangular faces. The opposite faces of a cuboid are identical.

In figure ABCD, EFGH, ADHE, BCGF, ABFE and DCGH are faces of cuboid.

(ii) Edge: The boundary of two adjacent faces of a cuboid meets in a line segment, which is called the edge of the cuboid.

A cuboid has 12 edges. In the figure, the edges are AB, BC, CD, DA, EF, GH, FG, EH, CG, BF, AE and DH.

(iii) Vertices: The point where three edges of a cuboid meet is called a vertex. A cuboid has 8 vertices. In the figure vertices are A, B, C, D, E, F, G and H.



Cuboid

(b) Cube

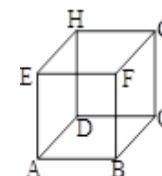
A cube is a cuboid whose all dimensions are equal. A cube is made up of 6 identical square faces.

Various parts of a cube are:

(i) Face: A cube has six square faces. In the figure, ABCD, EFGH, BCGF, ADHE, ABFE, and DCGH are faces.

(ii) Edge: Similar to a cuboid, a cube has 12 edges. In the figure, 12 edges are AB, BC, CD, DA, EF, GH, FG, EH, CG, BF, AE and DH.

(iii) Vertices: Similar to a cuboid, a cube has 8 vertices in total and here they are A, B, C, D, E, F, G, and H.



Cube

(c) Cylinder

A solid shape in which the top and bottom both are circular and joined by a curved surface.

Parts of a cylinder:

(i) A cylinder has no vertex.

(ii) A cylinder has two curved edges which are the circumferences of the circular faces.

(iii) A cylinder has one curved face and two flat faces.



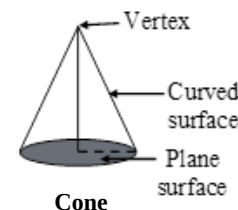
Cylinder

(d) Cone

A cone is a solid shape in which there is a plane circular base and a vertex which is connected to the base by a curved surface.

Parts of a cone:

(i) A cone has one vertex.



- (ii) A cone has one curved edge which is the circumference of the circular base.
 (iii) A cone has one curved surface and one flat or plane surface.

(e) Sphere

A solid round shape like a ball that has only a curved surface is called a sphere.

Parts of a sphere:

- (i) A sphere has no vertex.
 (ii) A sphere has no edge.
 (iii) A sphere only has a curved surface and no flat surfaces.



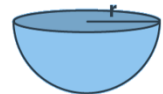
Sphere

(f) Hemisphere

A solid shape that is half of a sphere with one flat circular surface and one curved surface as in a circular bowl.

Parts of a sphere:

- (i) A sphere has no vertex.
 (ii) A sphere has one circular edge.
 (iii) A sphere only has a curved surface and 1 flat surface.

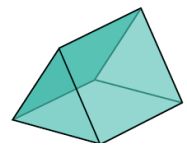


Hemisphere

(g) Prism

A prism is a solid with identical ends, flat surfaces, and the same cross-sectional area all along its length. The top and the bottom surfaces are two congruent and parallel polygons (i.e., base). The lateral faces are parallelograms.

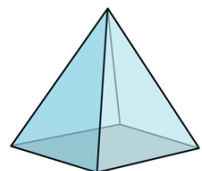
If two parallel and congruent polygons are regular and if the axis is perpendicular to the base, then the prism is called a right prism. A prism is named after the shape of these bases. For example, if a prism has a triangular base it is called a triangular prism. The lateral surfaces of a right prism are rectangles. Cuboids and cubes are also a type of prism.



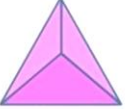
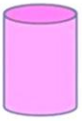
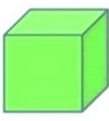
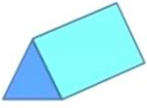
Prism

(h) Pyramid

A pyramid is a solid shape having a flat base (rectilinear) and triangular side faces which all meet at a common point called a vertex of the pyramid. Pyramids can be of various types depending upon the base of the pyramid like triangular pyramids known as tetrahedron, square pyramids, etc.



Pyramid

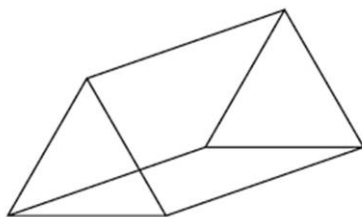
Properties of 3D shapes			
Cone  2 Faces 1 Edge 1 Vertex	Sphere  1 Face 0 Edge 0 Vertices	Tetrahedron  4 Faces 6 Edges 4 Vertices	Cuboid  6 Faces 12 Edges 8 Vertices
Cylinder  3 Faces 2 Edges 0 Vertices	Cube  6 Faces 12 Edges 8 Vertices	Triangular Prism  5 Faces 9 Edges 6 Vertices	Square-based pyramid  5 Faces 8 Edges 5 Vertices

Examples

- (i) Count how many faces, vertices and edges are there in the given solids shapes.



(a)



(b)

Solution: (a) It has 3 faces = 1 curved surface and 2 flat circular surfaces (i.e., top and bottom)
It has 0 vertices and it has 2 circular edges.
(b) It has 5 faces, 6 vertices and 9 edges.

12.3 Euler's Formula

The above figure 1 clearly shows that Faces + Vertices = Edges + 2

Leonhard Euler (1707 – 1783) discovered this formula which established the relationship among the number of faces, edges and vertices of a polyhedron.

$$\text{Euler's Formula: } F + V = E + 2$$

where, F = number of faces

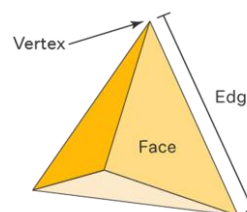
V = number of vertices

E = number of edges

Examples

- (i) Verify the Euler's formula for the given polyhedron.

Solution: No. of faces = $F = 4$
No. of Vertices = $V = 4$
No. of Edges = $E = 6$
Euler's formula: $F + V = E + 2$
 $\Rightarrow 4 + 4 = 6 + 2$
 $\Rightarrow 8 = 8$
LHS = RHS (Hence Proved)



- (ii) Prove the Euler's formula for a triangular prism.

Solution: Number of vertices of a triangular prism = 6
Number of edges of a triangular prism = 9
Number of faces of a triangular prism = 5
By the Euler's formula
 $V - E + F = 6 - 9 + 5 = -3 + 5 = 2$
 $\Rightarrow V - E + F = 2$
Thus, Euler's formula is satisfied for a triangular prism.

(iii) Is cuboid a polyhedron?

Solution: If cuboid is a polyhedron then it must satisfy Euler's formula for polyhedron

Number of edges, $e = 12$

Number of vertices, $v = 8$

Number of faces, $f = 6$

By the Euler's formula:

$$v - e + f = 8 - 12 + 6 = -4 + 6 = 2.$$

$$\Rightarrow v - e + f = 2.$$

which satisfies the Euler's formula. Hence, the cuboid is a polyhedron.

12.4 Nets

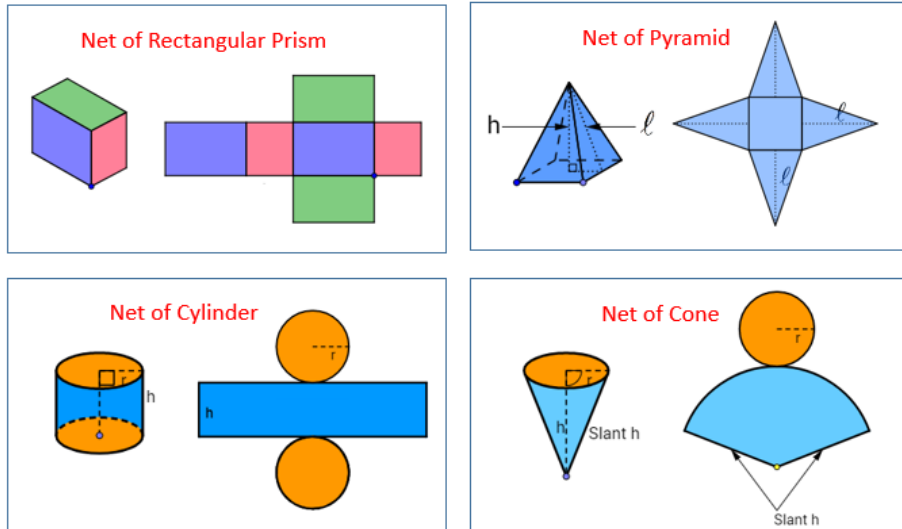
A net is a 2-dimensional shape that can be folded to form a 3-dimensional shape or a solid.

Here are some steps to determine whether a net forms a solid:

- Make sure that the solid and the net have the same number of faces and that the shapes of the faces of the solid match the shapes of the corresponding faces in the net.
- Visualize how the net is to be folded to form the solid and make sure that all the sides fit together properly. Nets are helpful when we need to find the surface area of the solids.

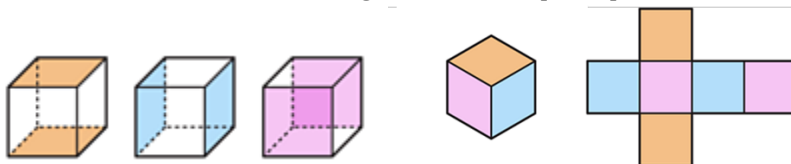
Net of some solid shapes are given below:

Nets of Solids



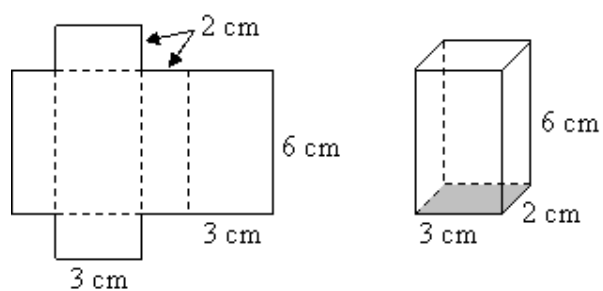
Net of A Cube

A cube is a three-dimensional figure with six equal square faces.



Net of a rectangular prism or cuboid

A rectangular prism or cuboid is formed by folding a net as shown:

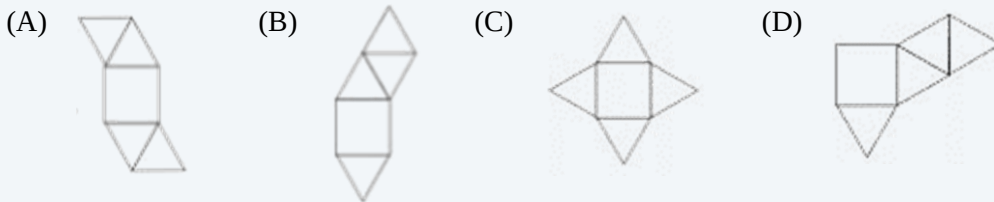




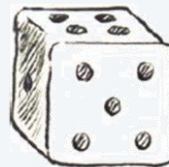
Practice Questions - 12

- (i) How many faces are there in a cuboid?
 (A) 1 (B) 2 (C) 4 (D) 6
- (ii) How many edges are there in a sphere?
 (A) 0 (B) 2 (C) 4 (D) 8
- (iii) How many flat faces are there in a cylinder?
 (A) 1 (B) 2 (C) 3 (D) 4
- (iv) If a polyhedron has 6 vertices and 12 edges. What is the number of faces it has?
 (A) 6 (B) 8 (C) 12 (D) 18
- (v) A polyhedron is said to be regular if its faces are made up of regular polygons and the same number of faces meet at each vertex.
 (A) True (B) False

- (vi) Which of the following options is not a net for a square-based pyramid?



- (vii) What is the number of vertices in a polyhedron with 30 edges and 12 faces?
- (viii) Dice are cubes with dots on each face. Opposite faces of a die always have a total of seven dots on them.

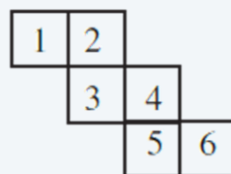


Here are two nets to make dice (cubes); the numbers inserted in each square indicate the number of dots in that box.

Insert suitable numbers in the blanks, remembering that the number on the opposite faces should total to 7.



- (ix) Can this be a net for a die?
 Explain your answer.



13. DATA HANDLING

13.1 Bar Graph

A bar graph is a pictorial representation of the numerical data (generally grouped) by a number of bars (rectangles) of uniform width erected horizontally or vertically with equal spacing between them.

Step 1: On graph paper draw a horizontal line and vertical line OX and OY respectively, which are called X-axis and Y-axis respectively.

Step 2: Mark points at equal spaces (intervals) along the X-axis. Below these points write the names of the data whose values are to be plotted.

Step 3: Assume (choose) a suitable scale. On that scale find the heights of the bars for the given numerical values.

Step 4: Mark these heights parallel to Y-axis from the point taken in step (ii).

Step 5: Draw bars of equal width for heights marked in step (iv) on X-axis. The bars should be centered on the points marked on X-axis. These bars show the given numerical data.

Examples

- (i) **Raju recorded the temperature at 2 p.m. for a week as follows: Draw a bar graph for the given data.**

Days	Mon	Tues	Wed	Thu	Fri	Sat	Sun
Temperature (°C)	30	25	35	30	20	38	28

Solution: We construct the bar graph using the following steps:

Take graph paper and draw two perpendicular lines OX and OY.

(1) Along OX mark "Names of the days" and along OY "Temperature in °C".

(2) Along OX, choose suitable width for each bar.

(3) Along OY, choose an appropriate scale and mark the temperature.

Here, 1 large division = 10° C

(4) Calculate the heights of various bars as follows:

(a) Height of bar for Monday = $\frac{1}{10} \times 30 = 3$ large divisions.

(b) Height of bar for Tuesday = $\frac{1}{10} \times 25 = 2.5$ large divisions.

(c) Height of bar for Wednesday = $\frac{1}{10} \times 35 = 3.5$ large divisions.

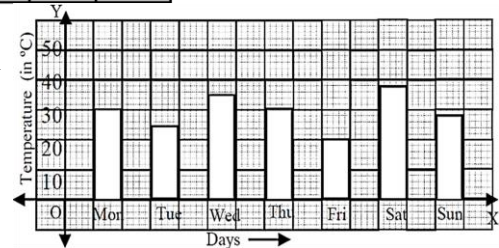
(d) Height of bar for Thursday = $\frac{1}{10} \times 30 = 3$ large divisions.

(e) Height of bar for Friday = $\frac{1}{10} \times 20 = 2$ large divisions.

(f) Height of bar for Saturday = $\frac{1}{10} \times 38 = 3.8$ large divisions.

(g) Height of bar for Sunday = $\frac{1}{10} \times 28 = 2.8$ large divisions.

And then, plot these bars on graph paper.



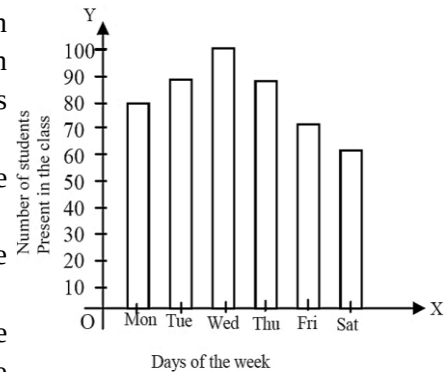
- (ii) Answer the following questions for the given bar graph:
- What does the bar graph represent?
 - On which day minimum number of students attended the class?
 - On which day was the highest attendance recorded?
 - If all students attended the class on Wednesday, how many students are there in the class?

Solution: (a) Along the horizontal line, the days of week are shown and along the vertical line, the students who were present on the days of the week are shown. So, the bar graph represents the attendance of a class during the days of a week.

(b) The bar for Saturday is the shortest, therefore, the minimum attendance was on Saturday.

(c) The bar for Wednesday is the highest, therefore, the maximum/highest attendance was on Wednesday.

(d) Take a scale/ruler and put it parallel to the line OX at the highest point of the bar for Wednesday. We find that the scale touches the OY line at 100 marks. So, the total number of students in the class are 100.



13.2 Central Tendency

Measures of central tendency are the values that describe a data set by identifying the central position of the data. There are 3 main measures of central tendency - Mean, Median and Mode.

(a) Mean

$$\text{Mean} = \frac{\text{Sum of the given observations}}{\text{Number of the given observations}}$$

Examples

- (i) The maximum temperature recorded in a city during the last six months of a year is given below.

Find the mean of the maximum temperature.

Month	July	Aug.	Sep.	Oct.	Nov.	Dec.
Maximum Temp. (in °C)	42	40	40	37	29	22

Solution: Total temp. = $42 + 40 + 40 + 37 + 29 + 22 = 210$

Number of months = 6

$$\text{Arithmetic Mean} = \frac{\text{Total Temperature}}{\text{Number of months}} = \frac{210}{6} = 35$$

Thus, the average or mean temperature = 35°C .

- (ii) **Given that mean of 5 numbers is 25. If the four numbers are 16, 26, 20, 32. Find the fifth number.**

Solution: Let the missing number be x .

The sum of the numbers = $16 + 26 + 20 + 32 + x = 94 + x$

Now Mean = $\frac{\text{Sum of numbers}}{5}$

$\therefore 5 \times \text{Mean} = \text{Sum of the numbers}$

i.e., $5 \times 25 = 94 + x$

or $125 = 94 + x \Rightarrow x = 125 - 94 = 31$

Thus, the fifth number is 31.

(b) Median

The value of the middlemost observation, obtained after arranging the data in ascending or descending order, is called the median of the data.

To find the median, we need to consider if n is even or odd.

(a) If n is odd, then use the formula:

$$\text{Median} = \left(\frac{n+1}{2} \right)^{\text{th}} \text{ observation}$$

(b) If n is even, then use the formula:

$$\text{Median} = \frac{\left[\left(\frac{n}{2} \right)^{\text{th}} \text{ observation} + \left(\left(\frac{n}{2} \right) + 1 \right)^{\text{th}} \text{ observation} \right]}{2}$$

Examples

- (i) **Find the median of the data: 34, 46, 56, 27, 28, 35, 45.**

Solution: First arranging in ascending order, we get 27, 28, 34, 35, 45, 46, 56

Here, $n = 7$ (which is odd)

$$\therefore \text{Median} = \left(\frac{7+1}{2} \right)^{\text{th}} \text{ term} = \left(\frac{8}{2} \right)^{\text{th}} \text{ term} = 4^{\text{th}} \text{ term} = 35$$

- (ii) **The weights of 8 students (in kg) are: 30, 42, 24, 37, 21, 25, 38, 31. Find median.**

Solution: Arranging the weights in ascending order, we get 21, 24, 25, 30, 31, 37, 38, 42

$N = 8$ (even)

$$\begin{aligned} \text{Median} &= \frac{\left(\frac{n}{2} \right)^{\text{th}} \text{ term} + \left(\left(\frac{n}{2} \right) + 1 \right)^{\text{th}} \text{ term}}{2} = \frac{\left(\frac{8}{2} \right)^{\text{th}} \text{ term} + \left(\left(\frac{8}{2} \right) + 1 \right)^{\text{th}} \text{ term}}{2} \\ &= \frac{(4^{\text{th}} + 5^{\text{th}}) \text{ term}}{2} = \frac{30+31}{2} = \frac{61}{2} = 30.5 \text{ kg.} \end{aligned}$$

(c) Mode

The mode is one of the values of the measures of central tendency. This value gives us a rough idea about which of the items in a data set tend to occur most frequently.

Examples

(i) Find the mode of numbers: 2, 2, 2, 2, 3, 3, 3, 4, 4, 5, 5, 5, 6, 6

Solution: Number 2 has the highest frequency (4), so the mode is 2.

(ii) The following table shows the marks obtained by 40 students of class VII. Find the mode.

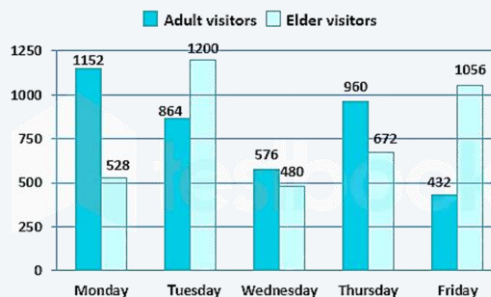
Marks Obtained	30	25	23	19	17	14
No. of Students	3	2	4	11	13	7

Solution: As the marks obtained by maximum number of students is 17 so the mode of the given data is 17.



Practice Questions - 13

- (i) The arithmetic mean of the first ten natural numbers is
 (A) 5.5 (B) 6
 (C) 7.5 (D) 10
- (ii) The median of the following data 46, 64, 87, 41, 58, 77, 35, 90, 5, 33, 92 is
 (A) 87 (B) 77
 (C) 58 (D) 60.2
- (iii) Find the mode of data 13, 16, 12, 14, 19, 12, 14, 13, 14
 (A) 12 (B) 13
 (C) 14 (D) 16
- (iv) The runs scored in a cricket match by 11 players are as follows:
 6, 15, 120, 50, 100, 80, 10, 15, 8, 10, 10
 Find the mean, mode and median of this data.
- (v) The scores in mathematics test (out of 25) of 15 students are as follows:
 19, 25, 23, 20, 9, 20, 15, 10, 5, 16, 25, 20, 24, 12, 20
 Find the mode and median of this data. Are they same?
- (vi) Answer the questions based on the information given below.

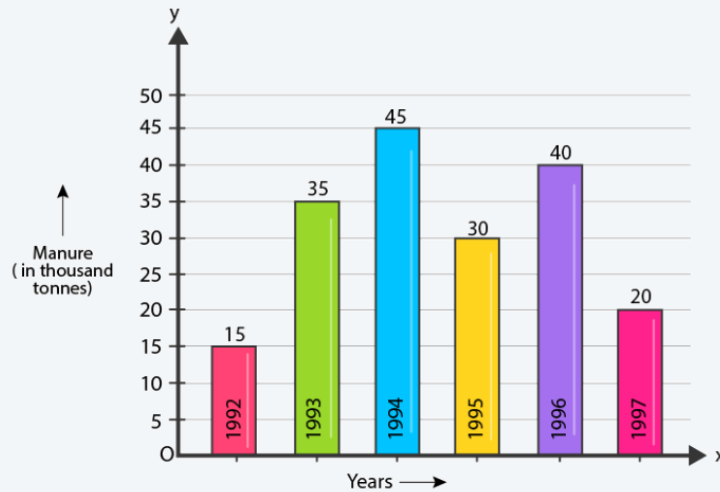


The bar graph given below shows the total number of visitors (Children + Adults + Elders) visiting a park during different days of the week. The total number of 'Children' who are visiting the park on any given day is the average of the number of 'Adults' and 'Elders' visiting the park on the same given day.

If the total number of adult' and 'Elder visitors visiting the park on Sunday is 25% less than that on Thursday respectively, then find the total number of visitors who are visiting the park on Sunday.

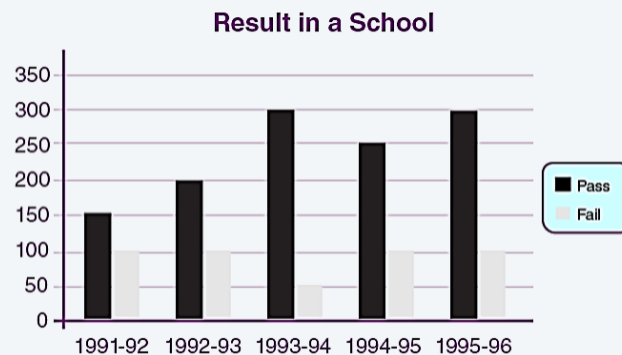
- (A) 1836 (B) 1846
 (C) 1856 (D) 1866

- (vii) The following data gives the amount of manure (in thousand tonnes) manufactured by a company during some years:



The consecutive years during which there was maximum decrease in manure production are:

- (A) 1994 and 1995 (B) 1992 and 1993
 (C) 1996 and 1997 (D) 1995 and 1996
- (viii) the year in which the amount of manure manufactured by the company was maximum.
- (ix) Examine the graph below carefully and answer the following questions. The graph depicts the results of a school's students.



Which year has the smallest difference between the number of kids who passed and those who failed?

- (x) In the last five years, what was the average number of kids who failed in school?

14. CHANCE & PROBABILITY

14.1 Important Terms

Term	Meaning	Example
Random Experiment	An experiment whose outcome cannot be predicted in advance	Tossing a coin, rolling a die
Outcome	A single possible result of a random experiment	Getting Head on a coin toss
Sample Space (S)	The set of all possible outcomes of an experiment	$S = \{H, T\}$ for a coin toss
Event (E)	A collection of one or more outcomes from the sample space	Getting an even number on a die
Favourable Outcomes	Outcomes that satisfy the condition of the event	Getting $\{2, 4, 6\}$ for "even number"
Equally Likely	When all outcomes have the same chance of occurring	Head and Tail on a fair coin

14.2 Probability

The number of favourable outcomes to the total number of outcomes is defined as the probability of occurrence of any event. So, the probability that an event E will occur is given as:

$$P(E) = \frac{\text{Number of Favourable Outcomes}}{\text{Total Number of Outcomes}}$$

14.3 Axioms of Probability

- (a) $0 \leq P(E) \leq 1$, for any event E.
- (b) Any probability must be a non-negative number.
- (c) $P(\text{certain event}) = 1$, $P(\text{sample space}) = 1$, $P(\text{impossible event}) = 0$.
- (d) In an experiment, the sum of the probability of all events is 1.
- (e) The sum of probabilities of occurring an event and not occurring is equal to 1.

Examples

- (i) **There are 6 balls in a box, 3 are red, 2 are yellow and 1 is blue. What is the probability of picking a yellow ball?**

Solution: The probability is equal to the number of yellow balls in the box divided by the total number of balls, i.e.,

$$\frac{2}{6} = \frac{1}{3}$$

(ii) There is a container full of coloured bottles, red, blue, green and orange. Some of the bottles are picked out and displaced. Riya did this 1000 times and got the following results:

- No. of blue bottles picked out: 200
- No. of red bottles: 300
- No. of green bottles: 50
- No. of orange bottles: 450

(a) What is the probability that Riya will pick a green bottle?

(b) If there are 100 bottles in the container, how many of them are likely to be green?

(c) What is the probability that Riya will pick a blue bottle?

Solution: (a) For every 1000 bottles picked out, 50 are green.

$$\text{Therefore, } P(\text{green}) = \frac{50}{1000} = 0.05$$

(b) The experiment implies that 50 out of 1000 bottles are green.

Therefore, out of 100 bottles, 5 are green.

(c) For every 1000 bottles picked out, 200 are blue.

$$\text{Therefore, } P(\text{blue}) = \frac{200}{1000} = 0.2$$

(iii) If today is Wednesday, then find the probability that tomorrow will be Thursday.

Solution: As it is a sure event, and the probability of occurrence of a sure event is always 1.

So, if today is Wednesday, then the probability that tomorrow will be Thursday is 1.

14.4 Sample Spaces for Common Experiments

(a) Tossing a Single Coin: $S = \{H, T\}$ $n(S) = 2$

(b) Tossing Two Coins: $S = \{HH, HT, TH, TT\}$ $n(S) = 4$

(c) Tossing Three Coins: $S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$ $n(S) = 8$

(d) Rolling a Single Die: $S = \{1, 2, 3, 4, 5, 6\}$ $n(S) = 6$

(e) Rolling Two Dice: $n(S) = 6 \times 6 = 36$ (all pairs from (1,1) to (6,6))

(f) Drawing a Card from a Standard Deck:

Category	Count	Details
Total Cards	52	
Suits	4	Hearts ♥, Diamonds ♦, Clubs ♣, Spades ♠
Cards per Suit	13	A, 2, 3, 4, 5, 6, 7, 8, 9, 10, J, Q, K
Red Cards	26	Hearts + Diamonds
Black Cards	26	Clubs + Spades
Face Cards (J, Q, K)	12	3 per suit $\times$ 4 suits
Aces	4	1 per suit $\times$ 4 suits
Number Cards (2–10)	36	9 per suit $\times$ 4 suits

14.5 Types of Events

Event Type	Probability	Example
Impossible Event	$P(E) = 0$	Getting 7 on a standard die
Sure / Certain Event	$P(E) = 1$	Getting a number ≤ 6 on a die
Equally Likely Events	$P = 1/n$ each	Head or Tail on a fair coin
Complementary Event E'	$P(E') = 1 - P(E)$	Not getting a 6 $= 1 - 1/6 = 5/6$

Examples

- (i) **Two fair coins are tossed. What is the probability that at least one head occurs?**

Solution: The sample space 'S' for this experiment is, $S = \{HH, HT, TH, TT\}$

If 'A' is the event representing the event of at least one head occurring,

$$\text{Then } A = \{HT, TH, HH\} \Rightarrow P(A) = \frac{3}{4}$$

- (ii) **From a set of 17 cards 1, 2, 3,16, 17, one is drawn at random. Find the probability that the number on the drawn card would be divisible by 3 or 7.**

Solution: Numbers which are divisible by three are 3, 6, 9, 12, and 15. Similarly, numbers which are divisible by 7 are 7, 14.

Total numbers which are divisible by 3 or 7 = 7 (3, 6, 7, 9, 12, 14, 15)

Probability of the number written on the card to be divisible by 3 or 7 is $= \frac{7}{17}$

- (iii) **A coin is tossed 500 times with the following frequencies of two outcomes:**

Head: 240 times, tail: 260 times. Find the probability of occurrence of each of these events.

Solution: It is given that the coin is tossed 500 times.

$\therefore$ Total number of trials = 500

Let us denote the event of getting a head and of getting a tail by A and B respectively. Then, number of trials in which the event A happens = 240.

and number of trials in which the event B happens = 260.

$$\therefore P(A) = \frac{\text{Number of trials in which event A happens}}{\text{Total number of trials}}$$

$$= \frac{240}{500} = 0.48$$

$$\therefore P(B) = \frac{\text{Number of trials in which the event B happens}}{\text{Total number of trials}}$$

$$= \frac{260}{500} = 0.52$$

Note: We note that $P(A) + P(B) = 0.48 + 0.52 = 1$. Therefore, A and B are the only two possible outcomes of trials.

- (iv) A card is drawn from a well-shuffled pack of 52 cards. Find the probability of getting
 (a) 2 of spades (b) a jack (c) a king of red colour
 (d) a card of diamond (e) a king or a queen

Solution: In a playing card there are 52 cards.

Therefore, the total number of possible outcomes = 52

(a) 2 of spades

Number of favourable outcomes i.e. '2 of spades' is 1 out of 52 cards.

Therefore, probability of getting '2 of spades' is

$$P(A) = \frac{\text{Number of favorable outcomes}}{\text{Total number of possible outcomes}} = \frac{1}{52}$$

(b) a jack

Number of favourable outcomes i.e. 'a jack' is 4 out of 52 cards.

Therefore, probability of getting 'a jack' is

$$P(B) = \frac{\text{Number of favorable outcomes}}{\text{Total number of possible outcomes}} = \frac{4}{52} = \frac{1}{13}$$

(c) a king of red colour

Number of favourable outcomes i.e. 'a king of red colour' is 2 out of 52 cards. Therefore, probability of getting 'a king of red colour' is

$$P(C) = \frac{\text{Number of favorable outcomes}}{\text{Total number of possible outcomes}} = \frac{2}{52} = \frac{1}{26}$$

(d) a card of diamond

Number of favourable outcomes i.e. 'a card of diamond' is 13 out of 52 cards. Therefore, probability of getting 'a card of diamond' is

$$P(D) = \frac{\text{Number of favorable outcomes}}{\text{Total number of possible outcomes}} = \frac{13}{52} = \frac{1}{4}$$

(e) a king or a queen

Total number of king is 4 out of 52 cards.

Total number of queen is 4 out of 52 cards.

Number of favourable outcomes i.e. 'a king or a queen' is $4 + 4 = 8$ out of 52 cards.

Therefore, probability of getting 'a king or a queen' is

$$P(E) = \frac{\text{Number of favorable outcomes}}{\text{Total number of possible outcomes}} = \frac{8}{52} = \frac{2}{13}$$



Practice Questions - 14

- (i) Find the probability of the sun setting in the North.
- (ii) In 2010, there was a survey of 200 people and it was found that 82 liked orange juice while 118 disliked it. From these people, one is chosen at random. What is the probability that the chosen people disliked orange juice?
- (iii) In a box there are 10 non-defective and 5 defective bulbs. If a bulb is selected at random from the box then find the probability that it will be defective.
- (iv) A number is chosen from 1 to 30. What is the probability it is divisible by both 3 and 5?
 (A) $\frac{1}{15}$ (B) $\frac{2}{15}$
 (C) $\frac{1}{10}$ (D) $\frac{1}{6}$
- (v) In a factory of 265 workers, 132 are married. Find the probability of selecting a worker who is not married.
- (vi) A fair coin is tossed 450 times and the outcomes were noted as: Head = 250, Tail = 200. Find the probability of the coin showing up
 (a) a head (b) a tail.
- (vii) A bag contains blue and white balls. The probability of not getting a white balls from the bag is 0.36. What is the probability of getting a white ball?
- (viii) A bag contains red and white balls. The probability of getting a red ball from the bag of balls is $\frac{1}{6}$. What is the probability of not getting a red ball?
- (ix) A box contains blue and green marbles. The probability of getting a green marble from the box of marbles is $\frac{3}{7}$. What is the probability of getting a blue marble?
- (x) A card is drawn for the deck of 52 cards. The event E so that card is not the ace for the hearts. The number for the outcomes favourable to E is
 (A) 4 (B) 13
 (C) 48 (D) 51

15. MIXED SOLVED PROBLEMS

- (i) **A wire in the form of a rectangle 18.7 cm long and 14.3 cm wide is reshaped and bent in the form of a circle. Find the radius of the circle so formed.**

Solution Given: Length of rectangular wire = 18.7 cm and Breadth of rectangular wire = 14.3 cm

According to the question, length of wire = perimeter of the rectangle

$$= 2(l + b) = 2 \times (18.7 + 14.3)$$

$$= 2(33) = 66 \text{ cm}$$

Let the wire bent in the form of the circle be of radius r cm. then we have,

$$\text{Circumference} = 66 \text{ cm}$$

$$2\pi r = 66$$

$$\Rightarrow 2 \times \left(\frac{22}{7}\right) \times r = 66$$

$$\Rightarrow \left(\frac{44}{7}\right) r = 66$$

$$\Rightarrow r = \frac{66 \times 7}{44}$$

$$\Rightarrow r = \frac{462}{44} = 10.5 \text{ cm}$$

- (ii) **The cost of fencing a square field at 40 paise per metre is 2000 Rs., find the cost of watering the field at 80 paise per 100 m².**

Solution: Total cost of fencing = Rs. 2000.

$$\text{Rate of fencing} = 40 \text{ paise/m} = \left(\frac{40}{100}\right) \text{ Rs. per m}$$

$$= \frac{2}{5} \text{ Rs. per m.}$$

$$\text{The perimeter of the field} = \frac{\text{Total Cost of fencing}}{\text{Rate per Meter}}$$

$$= \frac{2000}{\frac{2}{5}} \text{ m} = \frac{2000 \times 5}{2} = 5,000 \text{ m}$$

$$\text{Side of the square field} = \frac{5000}{4} \text{ m} = 1250 \text{ m}$$

$$\text{Area of square field} = (1,250 \times 1,250) \text{ m}^2$$

$$\text{Area of square field} = 15,62,500 \text{ m}^2$$

$$\text{Cost of watering } 100 \text{ m}^2 = 80 \text{ paise} = \left(\frac{80}{100}\right) = \frac{4}{5}$$

$$\text{Cost of watering } 1 \text{ m}^2 = \left(\frac{4}{5} \times \frac{1}{100}\right)$$

$$\text{Cost of watering } 15,62,500 \text{ m}^2$$

$$= \left(\frac{4}{5} \times \frac{1}{100} \times 1562500\right) = \text{Rs. } 12,500.$$

(iii) In the following figure, find the unknown angles a and b , if $l \parallel m$.

Solution: Here, $l \parallel m$, $\angle c = 110^\circ$ [Corresponding angles]

$$\angle c + \angle a = 180^\circ \quad [\text{Linear pair}]$$

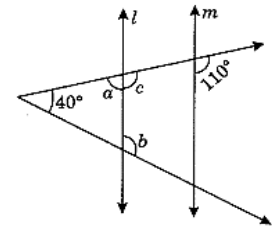
$$\Rightarrow 110^\circ + \angle a = 180^\circ$$

$$\Rightarrow \angle a = 180^\circ - 110^\circ = 70^\circ$$

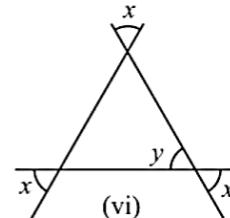
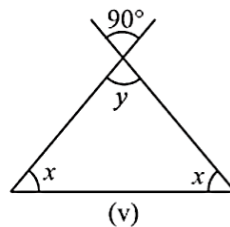
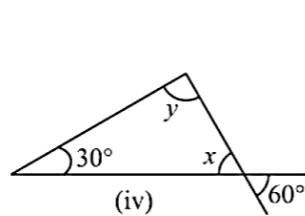
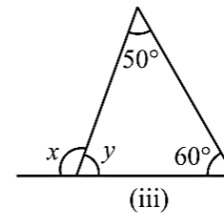
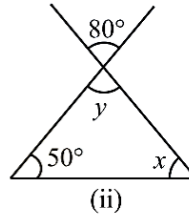
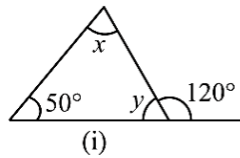
Now $\angle b = 40^\circ + \angle a$ [Exterior angle of a triangle]

$$\Rightarrow \angle b = 40^\circ + 70^\circ = 110^\circ$$

Hence, the values of unknown angles are $a = 70^\circ$ and $b = 110^\circ$.



(iv) Find the values of the unknowns x and y in the following diagrams:



Solution: (i) We know that, Sum of interior opposite angles of a triangle = Exterior angle

$$50^\circ + x = 120^\circ$$

$$x = 120^\circ - 50^\circ$$

$$x = 70^\circ$$

By angle sum property, Sum of interior angles = 180°

$$50^\circ + 70^\circ + y = 180^\circ$$

$$120^\circ + y = 180^\circ$$

$$y = 180^\circ - 120^\circ$$

$$y = 60^\circ$$

(ii) $y = 80^\circ$ (vertically opposite angles)

By angle sum property Sum of interior angles = 180°

$$50^\circ + x + y = 180^\circ$$

$$50^\circ + x + 80^\circ = 180^\circ$$

$$130^\circ + x = 180^\circ$$

$$x = 180^\circ - 130^\circ$$

$$x = 50^\circ$$

(iii) Solve for x and y

We know that, Sum of interior opposite angles of a triangle = Exterior angle

$$50^\circ + 60^\circ = x$$

$$110^\circ = x$$

$$x = 110^\circ$$

By angle sum property,

$$\text{Sum of interior angles} = 180^\circ$$

$$50^\circ + 60^\circ + y = 180^\circ$$

$$110^\circ + y = 180^\circ$$

$$y = 180^\circ - 110^\circ$$

$$y = 70^\circ$$

(iv) Solve for x and y

$$x = 60^\circ$$

(vertically opposite angles are equal)

We know that,

Sum of interior angles of a triangle = 180°

$$30^\circ + x + y = 180^\circ$$

$$30^\circ + 60^\circ + y = 180^\circ$$

$$90^\circ + y = 180^\circ$$

$$y = 180^\circ - 90^\circ$$

$$y = 90^\circ$$

(v) Solve for x and y

$$y = 90^\circ$$

(vertically opposite angles are equal)

We know that, Sum of interior angles of a triangle = 180°

$$y + x + x = 180^\circ$$

$$90^\circ + 2x = 180^\circ$$

$$2x = 180^\circ - 90^\circ$$

$$x = \frac{90^\circ}{2}$$

$$x = 45^\circ$$

(vi) Solve for x and y

$$y = x$$

(vertically opposite angles are equal)

We know that, Sum of interior angles of a triangle = 180°

$$y + x + x = 180^\circ$$

$$x + x + x = 180^\circ$$

$$3x = 180^\circ$$

$$x = 60^\circ$$

Hence, $x = 60^\circ, y = 60^\circ$

(v) **If $p \parallel q$, then find the value of $\angle 1$, $\angle 2$, and $\angle 1 + \angle 2$.**

Solution: Since $p \parallel q$ and l is a transversal

$\angle PAC = \angle ECA$ (Alternate interior angles)

$$\Rightarrow \angle ECA = 139^\circ$$

$$\Rightarrow \angle ECB + \angle BCA = 139^\circ \Rightarrow 49^\circ + \angle BCA = 139^\circ$$

$$\Rightarrow \angle BCA = 139^\circ - 49^\circ \Rightarrow \angle BCA = 90^\circ$$

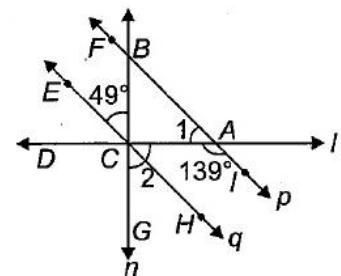
Now, $\angle BCA + \angle ACG = 180^\circ$ (linear pair)

$$\Rightarrow 90^\circ + \angle 2 = 180^\circ \Rightarrow \angle 2 = 180^\circ - 90^\circ \Rightarrow \angle 2 = 90^\circ$$

Also, $\angle DCE + \angle ECB + \angle BCA = 180^\circ$

(Angles on a straight line)

$$\Rightarrow \angle DCE + 49^\circ + 90^\circ = 180^\circ$$



$$\Rightarrow \angle DCE = 180^\circ - 139^\circ = 41^\circ$$

Now, $p \parallel q$ and l is a transversal

$$\therefore \angle BAC = \angle ECD \quad (\text{Corresponding angles})$$

$$\Rightarrow \angle 1 = 41^\circ. \text{ Hence, } \angle 1 + \angle 2 = 41^\circ + 90^\circ = 131^\circ$$

(vi) (a) If $x + y = 7$ and $xy = 2$, then find $x^2 - y^2$ and $(x > y)$

(b) If $2y + \frac{1}{2y} = l$ and $2y - \frac{1}{2y} = m$, then $l^2 - m^2 =$

(c) If $2y + \frac{1}{2y} = 3$, then $16y^4 + \frac{1}{16y^4} =$

Solution: (a) Given, $x + y = 7$, and $xy = 2$

$$\therefore (x - y)^2 = (x + y)^2 - 4xy$$

$$\Rightarrow x - y = \sqrt{(x + y)^2 - 4xy}$$

$$= \sqrt{7^2 - 4(2)} = \sqrt{41}$$

$$\text{Now, } x^2 - y^2 = (x + y)(x - y)$$

$$= 7\sqrt{41}$$

$$(b) 2y + \frac{1}{2y} = l$$

$$l^2 = (2y)^2 + 2 \times 2y \times \frac{1}{2y} + \left(\frac{1}{2y}\right)^2$$

$$l^2 = 4y^2 + \frac{1}{4y^2} + 2 \quad (1)$$

$$\& 2y - \frac{1}{2y} = m$$

$$\Rightarrow \left(2y - \frac{1}{2y}\right)^2 = m^2$$

$$m^2 = 4y^2 + \frac{1}{4y^2} - 2 \quad (2)$$

$$l^2 - m^2 = (1) - (2) = \left(4y^2 + \frac{1}{4y^2} + 2\right) - \left(4y^2 + \frac{1}{4y^2} - 2\right) = 4.$$

$$(c) \text{ Given, } 2y + \frac{1}{2y} = 3$$

Taking squares on both sides,

$$\Rightarrow \left(2y + \frac{1}{2y}\right)^2 = 3^2$$

$$\Rightarrow 4y^2 + \frac{1}{4y^2} + 2 = 9$$

$$\Rightarrow 4y^2 + \frac{1}{4y^2} = 7$$

Again, taking squares on both sides,

$$\Rightarrow \left(4y^2 + \frac{1}{4y^2}\right)^2 = 7^2$$

$$\Rightarrow 16y^4 + \frac{1}{16y^4} + 2 = 49$$

$$\Rightarrow 16y^4 + \frac{1}{16y^4} = 47$$

- (vii) (a) A grain merchant sold 600 quintals of rice at a profit of 7%. If one quintal of rice cost him ₹ 250 and his total overhead charges for transportation were ₹ 1000, find his total profit and the selling price of 600 quintals of rice.
 (b) Two tailors X and Y are paid a total of Rs. 550 per week by their employer. If X is paid 120 percent of the sum paid to Y, how much is Y paid per week?

Solution:

(a) Cost of 1 quintal of rice = ₹ 250
 Cost of 600 quintals of rice = $600 \times 250 = ₹ 150000$
 Overhead expenses = ₹ 1000
 Total CP = ₹ $(1,50,000 + 1000) = ₹ 151000$
 $\text{Profit \%} = \frac{\text{Profit}}{\text{CP}} \times 100$
 $7 = \frac{\text{Profit}}{151000} \times 100$
 $\text{Profit} = 1510 \times 7 = ₹ 10570$
 $\text{Profit} = ₹ 10570$
 $\text{SP} = \text{CP} + \text{profit} = ₹ (151000 + 10570) = ₹ 161570$
 (b) Let the sum paid to Y per week be Rs. z.
 Then, $z + 120\% \text{ of } z = 550$
 $\Rightarrow z + \frac{120}{100} z = 550$
 $\Rightarrow \frac{11}{5} z = 550$
 $\Rightarrow z = \left(\frac{550 \times 5}{11} \right) = 250$

- (viii) (a) Evaluate: $\frac{15^4 \times 18^3}{3^3 \times 5^2 \times 12^2}$
 (b) Find the value of x, if $\frac{2^{2x} \times 4 \times 2^x - 8^x}{(2^5)^3 \times 9} = \frac{1}{24}$.

Solution:

(a) $\frac{(3 \times 5)^4 \times (2 \times 3 \times 3)^3}{3^3 \times 5^2 \times (2 \times 2 \times 3)^2} = \frac{3^4 \times 5^4 \times (2 \times 3^2)^3}{3^3 \times 5^2 \times (2^2 \times 3)^2}$
 $= \frac{3^4 \times 5^4 \times 2^3 \times (3^2)^3}{3^3 \times 5^2 \times (2^2)^2 \times 3^2} = \frac{3^4 \times 5^4 \times 2^3 \times 3^6}{3^3 \times 5^2 \times 2^4 \times 3^2}$
 $= \frac{3^{4+6} \times 5^4 \times 2^3}{3^{3+2} \times 5^2 \times 2^4} = \frac{3^{10} \times 5^4 \times 2^3}{3^5 \times 5^2 \times 2^4}$
 $= \frac{3^{10-5} \times 5^{4-2} \times 2^3}{2^{4-3}} = \frac{3^5 \times 5^2}{2}$
 $= \frac{243 \times 25}{2} = \frac{6075}{2}$
 $= 3037.50$
 (b) $\frac{2^{2x} \times 2^2 \times 2^x - (2^3)^x}{2^{5 \times 3} \times 3^2} = \frac{1}{24}$
 $\Rightarrow \frac{2^{2x+2+x} - 2^{3x}}{2^{15} \times 3^2} = \frac{1}{24} \Rightarrow \frac{2^{3x+2} - 2^{3x}}{2^{15} \times 3^2} = \frac{1}{24}$

$$\Rightarrow \frac{2^{3x}(2^2 - 1)}{2^{15} \times 3^2} = \frac{1}{24} \Rightarrow \frac{2^{3x-15} \times 3}{3^2} = \frac{1}{24}$$

$$\Rightarrow 2^{3x-15} = \frac{1}{8} \Rightarrow 2^{3x-15} = 2^{-3}$$

Comparing the powers of the same bases, we have $3x - 15 = -3 \Rightarrow 3x = 12 \Rightarrow x = 4$
Thus, $x = 4$.

(ix) (1) Express each of the following fractions as percentages:

(a) $\left(\frac{3}{4}\right)$ (b) $\left(\frac{53}{100}\right)$
(c) $1\frac{3}{5}$ (d) $\left(\frac{7}{20}\right)$

(2) Express each of the following percentages as fractions in the simplest forms:

(a) 45% (b) 0.25%
(c) 150% (d) $6\frac{1}{4}\%$

(3) If $(3a + 5b) : (3a - 5b) = 5 : 1$, then $a : b = ?$

(4) If $a : b = 4 : 5$ and $b : c = 2 : 3$, then $a : c = ?$

Solution:

(1)

(a) $\frac{3}{4} = \left(\frac{3}{4} \times 100\right)\% = 75\%$
(b) $\frac{53}{100} = \left(\frac{53}{100} \times 100\right)\% = 53\%$
(c) $1\frac{3}{5} = \left(\frac{8}{5} \times 100\right)\% = 160\%$
(d) $\frac{7}{20} = \left(\frac{7}{20} \times 100\right)\% = 35\%$

(2)

(a) $45\% = \frac{45}{100} = \frac{9}{20}$
(b) $0.25\% = \frac{0.25}{100} = \frac{25}{10000} = \frac{1}{400}$
(c) $150\% = \frac{150}{100} = \frac{3}{2}$
(d) $6\frac{1}{4}\% = \frac{6.25}{100} = \frac{625}{10000} = \frac{1}{16}$

(3) $\frac{3a + 5b}{3a - 5b} = \frac{5}{1}$

$$3a + 5b = 15a - 25b$$

$$12a = 30b$$

$$\frac{a}{b} = \frac{30}{12} = \frac{5}{2}$$

$$\therefore a : b = 5 : 2$$

(4) Given: $a : b = 4 : 5$

$$\Rightarrow \frac{a}{b} = \frac{4}{5}$$

Also, $b : c = 2 : 3$

$$\Rightarrow \frac{b}{c} = \frac{2}{3}$$

$$\text{So, } a : c = \frac{a}{c} = \frac{ab}{bc}$$

$$\Rightarrow \frac{a}{b} \times \frac{b}{c}$$

$$\Rightarrow \frac{4}{5} \times \frac{2}{3} \Rightarrow \frac{8}{15}$$

$$= 8 : 15$$

(x) (a) If the length of each edge is 6 cm in the given figure, then find the length of the tape required to cover each edge once.

(b) The given diagram shows the net of a solid. The perimeter of the whole shaded diagram is 56 cm. Calculate the length of the edge on the solid.



Solution: (a) The figure shown here is a triangular prism.

It is given that all the edges are of equal length.

As we can see here that there are 9 edges in this figure in total so we can conclude that the length of the required tape is equal to the perimeter of the given figure.

The perimeter of the given figure

= Number of edges $\times$ length of each edge

(since all the edges are equal)

$$= 9 \times 6 = 54 \text{ cm.}$$

(b) Let us take each square's side as x units.

Then, the perimeter of the shaded figure will be 14x, as 14 sides are included in the perimeter.

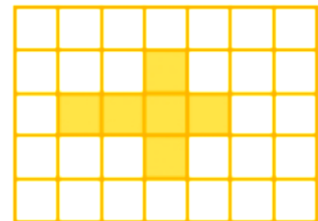
Now, it is given that the perimeter is 56 cm, thus

$$14x = 56$$

$$\Rightarrow x = \frac{56}{14}$$

$$\Rightarrow x = 4$$

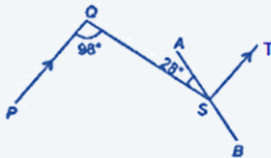
Thus, the length of the edge of the solid is 4 cm.



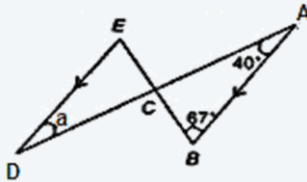
Practice Questions - 15



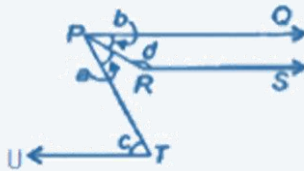
- (i) The area of a square park is the same as that of a rectangular park. If the side of the square park is 60 m and the length of the rectangular park is 90 m, find the breadth of the rectangular park.
- (ii) In a triangle, if the second angle is 5° greater than the first angle and the third angle is 5° greater than second angle, find the three angles of the triangle.
- (iii) (a) In the figure, PQ is parallel to ST. AB is a straight line. Find $\angle BST$.



- (b) In the given figure, $AB \parallel DE$. Find a .



- (c) In the given figure, PQ, RS, and UT are parallel lines. If $c = 75^\circ$ and $a = \left(\frac{2}{5}\right)c$, find $b + \frac{d}{2}$.



- (iv) (a) From the sum of $2p^2 + 3pq$, $-p^2 - pq - q^2$ and $pq + 2q^2$ subtract the sum of $3p^2 - q^2$ and $-p^2 + pq + q^2$.
- (b) Subtract the sum of $7x^2 - 4y^2$ and $4x^2 + 3xy - 4y^2$ from the sum of $5x^2 - 3xy - y^2$ and $x^2 + 2xy - 2y^2$.
- (c) Add: $5x^2 - \frac{1}{3}x + \frac{5}{2} - \frac{1}{2}x^2 + \frac{1}{2}x - \frac{1}{3}$ and $-2x^2 + \frac{1}{5}x - \frac{1}{6}$
- (v) Given the following values, find the unknown values:
- (a) C.P. = ₹ 1200, S.P. = ₹ 1350, Profit/Loss = ?
- (b) C.P. = ₹ 980, S.P. = ₹ 940, Profit/Loss = ?
- (c) C.P. = ₹ 720, S.P. = ? Profit = ₹ 55.50
- (d) C.P. = ? S.P. = ₹ 1254, Loss = ₹ 32

- (vi) (a) A sum of Rs 1750 is divided into two parts such that the interests on the first part at 8% simple interest per annum and that on the other part at 6% simple interest per annum are equal. What is the interest accumulated on each part (in Rs)?
 (b) A sum of money amounts to Rs. 9800 after 5 years and Rs. 12005 after 8 years at the same rate of simple interest. Find out the rate of interest per annum.
- (vii) (a) Find the fourth proportional to $x + 3x$, $x + 2$ and $5x$, if $x = 3$.
 (b) Find the third proportional to 9 and 15.
 (c) Find the mean proportional between 12.5 and 50.
- (viii) (a) In a stack there are 5 books each of thickness 20 mm and 5 paper sheets each of thickness 0.016 mm. What is the total thickness of the stack?
 (b) Weight of moon is (7.346×10^{22}) kg and weight of Earth is 5.9724×10^{24} kg. What is the total weight of both in the standard form?
 (c) Write the following numbers in the standard form:
 (1) 0.000000564 (2) 0.00000000000942
- (ix) (a) A cricketer hit 120 runs in 150 balls during a test match. 20% of the runs came in 6's, 30% in 4's, 25% in 2's and the rest in 1's. How many runs did he score in
 (i) 6's
 (ii) 4's
 (iii) 2's
 (iv) singles
 What % of his shots were scoring ones?
 (b) Gunpowder contains 75% nitre and 10% sulphur. Find the amount of the gunpowder which carries 9 kg nitre. What amount of gunpowder would contain 2.3 kg sulphur?
- (x) (a) A rectangular park of length 30 m and breadth 25 m has a path of uniform width 5 m around its perimeter. Find the cost of tiling the whole path if the cost of tiling per 10 m^2 is ₹ 1500.
 (b) The floor of a room has to be covered with square tiles 30 cm each. If the length and the breadth of the room are 22.5 m and 10 m, respectively, find how many tiles are required to cover the whole floor.
 (c) The perimeter of a square of side 25 cm is equal to the perimeter of a rectangle whose width is $\frac{2}{5}$ th the side of the square. Find the area of the rectangle.

Practice Questions - Answer Key

- (1) (i) (a) 44 (b) 32 (c) -54 (d) -80 (ii) 33 (iii) (a) 4 (b) -40 (iv) (B)
 (v) First Part (vi) (C) (vii) 60
- (2) (i) (D) (ii) (C) (iii) (A) (iv) (A) (v) (D)
 (vi) (D) (vii) (B) (viii) (C) (ix) No (x) No
- (3) (i) (C) (ii) (C) (iii) (C) (iv) (B) (v) (C)
 (vi) (A) (vii) (A) (viii) (a) $3.49 \times 10^4 = 34,900$ (b) $1.11 \times 10^6 = 1,110,000$
 (ix) 4.1763×10^9 (x) Yes (xi) 31 km (xii) (a) 4750 m (b) 83.5 km (c) 2300 g
 (xiii) 5000 milligrams (xiv) 26 kg 700 g (xv) 6 kg 50 g
- (4) (i) $-5x^2 + 7x + 9$ (ii) 0 (iii) $3x^3 - x^2 + x - 1$ (iv) (A)
 (v) (D) (vi) (C) (vii) 6,400 (ix) (a) 9,98,001 (b) 4,94,209
 (viii) (a) $4x^2 + 4xy + y^2$ (b) $a^2 - 4b^2$
 (x) (a) 16×24 is larger (b) 26×74 is larger
- (5) (i) (D) (ii) (A) (iii) (A) (iv) (A) (v) (D)
 (vi) $38\%, \frac{62}{100}, \frac{24}{100}, 600$ (vii) 24 years (viii) Greater = 112 (ix) $m = -13$
 (x) 36 years
- (6) (i) (B) (ii) (D) (iii) (D) (iv) (B) (v) (D)
 (vi) (A) (vii) (B) (viii) (D) (ix) (C) (x) (B)
- (7) (i) $\frac{1}{3}$ (ii) (a) 25% (b) 50% (c) 16% (iii) (a) $\frac{9}{20}$ (b) $\frac{13}{20}$ (c) $\frac{17}{40}$
 (iv) 7 Km, 3 Km (v) 6 kg (vi) $4\frac{11}{12}$ (vii) 2 hours 30 minutes
 (viii) Rs. 547200 (ix) 6.25% (x) Rs. 72
- (8) (i) Rs. 2750 Profit (ii) 50% Profit (iii) 46% Profit (iv) 6% (v) 8.33%
 (vi) (C) (vii) (B) (viii) (A) (ix) (D) (x) (C)
- (9) (i) Supplementary (ii) 51° and 39° (iii) 90° (iv) 130° (v) $\angle PLB = 60^\circ$ (vi) 90° (vii) 45° (viii) 108°
- (ix) (i) $\angle PAB$ and $\angle XAQ$ are vertically opposite angles.
 (ii) $\angle PAB$ and $\angle ABS$ are alternate angles.
 (iii) $\angle PAB$ and $\angle RBY$ are corresponding angles.
 (x) (A)
- (10) (i) (D) (ii) (B) (iii) (B) (iv) (A) (v) $\angle 1 = 105^\circ$
 (vi) 29 cm (vii) (C) (viii) (C) (ix) (C) (x) (C)
- (11) (i) (A) (ii) (A) (iii) (B) (iv) (B) (v) (B)
 (vi) (A) (vii) (C) (viii) Rs. 314 (ix) 96 cm^2 (x) 196 m^2
- (12) (i) (D) (ii) (A) (iii) (B) (iv) (B) (v) True
 (vi) (B) (vii) 20 (ix) No
- (13) (i) (A) (ii) (C) (iii) (C) (iv) The mean is approximately 38.55, The Mode is 10 and the median is 15
 (v) Yes, Median = Mode = 20
 (vi) (A) (vii) (C) (viii) 1994 (ix) 1991 - 1992 (x) 90

- (14) (i) 0 (ii) $\frac{59}{100}$ (iii) $\frac{1}{3}$ (iv) (A) (v) $\frac{133}{265}$
 (vi) (a) $\frac{5}{9}$ (b) $\frac{4}{9}$ (vii) 0.64 (viii) $\frac{5}{6}$ (ix) $\frac{4}{7}$ (x) (D)
- (15) (i) 40 m (ii) $55^\circ, 60^\circ, 65^\circ$ (iii) (a) 110° (b) 40° (c) 112.5°
 (iv) (a) $-p^2 + 2pq + q^2$ (b) $-5x^2 + 5y^2 - 4xy$ (c) $\frac{5x^2}{2} + \frac{11}{30}x + 2$
 (v) (a) ₹ 150 (b) ₹ 40 (c) ₹ 775.50 (d) ₹ 1286 (vi) (a) ₹ 60 (b) 12 %
 (vii) (a) 6.25 (b) 25 (c) 25
 (viii) (a) 1.0008×10^2 mm (b) 6.04586×10^{24} kg (c) (1) 5.64×10^{-7} (2) 9.42×10^{-12}
 (ix) (a) (i) 24 (ii) 36 (iii) 30 (iv) 25%
 (b) nitre is 12kg and sulphur is 23kg (x) (a) ₹ 97,500. (b) 2500 (c) 400 cm^2